Model-Agnostic FDR Control via Group Gaussian Mirror and Permutation SHAP

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Abstract

Most FDR-controlled feature selection methods are designed for coordinate-wise hypotheses, where each feature has a single weight or importance score. This abstraction fails in sequential and grouped models, where one original feature is represented by a block of sub-features, such as lags, recurrent states, or attention-based interactions. We propose a grouped-feature FDR control framework for such settings. For grouped linear models, we construct null-symmetric block-level mirror statistics with matrix-valued perturbations. For neural sequential models, we combine Permutation SHAP derivatives as model-agnostic block-level importance scores with kernel-based dependence measure. The framework is model-agnostic across network architectures, does not require specifying the covariate distribution, and reduces to Gaussian Mirror or Neural Gaussian Mirror when the block size is one. We prove FDR control for low- and high-dimensional grouped linear models and asymptotic symmetry of smoothed Permutation SHAP derivatives under fixed fitted nonlinear models. Experiments on simulated and real-world datasets show reliable FDR control and improved power under correlated grouped-feature signals. Code is available at: `https://github.com/HJA321/Model_Agnostic_FDR_Control`.

1 Introduction

Controlling the false discovery rate (FDR) is a central problem in feature selection, particularly when prediction and identifying truly relevant input variables are both important. In high-dimensional inference, selecting features without controlling FDR can lead to misleading scientific conclusions and unstable models. A large body of work has developed FDR-controlled procedures for feature selection, including the BH procedure [6], Knockoff-based methods [3, 9, 10, 27, 37, 65, 66, 49], Gaussian Mirror and Neural Gaussian Mirror [60, 59], and Data Splitting [13, 14, 48]. Among these, Gaussian Mirror is particularly appealing for not requiring a specified joint distribution of the covariates, unlike Knockoffs, and has higher statistical power in general settings [34].

Despite their differences, most existing FDR-controlled feature selection methods are built around a common prototype: each feature is associated with a single model weight. In the classical linear model $y = X\beta + \varepsilon$, feature-level inference is typically formulated through coordinate-wise hypotheses

$$H_{0,j} : \beta_j = 0, \qquad j = 1, \dots, p.$$

This single-weight formulation underlies Gaussian Mirror, Knockoff-based procedures, and many related methods. Even in nonlinear settings [14, 27, 65, 66], feature importance is often reduced to a scalar score per feature, preserving this one-feature-one-weight paradigm.

However, the one-feature-one-weight prototype is not the right abstraction for many grouped or sequential prediction problems. A clear example is a lagged time-series model $Y_t = \sum_{\ell=0}^{m-1} X_{t-\ell}^\top \boldsymbol{\beta}_\ell +$

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ε_t, where each $X_{t-\ell} \in \mathbb{R}^p$ contains the same p original features observed at lag ℓ. For an original feature j, the model contains a lag-weight vector $\boldsymbol{\beta}_j = (\beta_{j,0}, \beta_{j,1}, \ldots, \beta_{j,m-1})$. Thus, the scientific question should not be whether a single lag-specific weight $\beta_{j,\ell}$ is zero, but whether feature j has any predictive contribution across the whole lag window, which formulates the new hypotheses:

$$H_{0,j} : \beta_{j,0} = \beta_{j,1} = \cdots = \beta_{j,m-1} = 0.$$

Equivalently, if $X_j = [\boldsymbol{x}_{j,t}, \boldsymbol{x}_{j,t-1}, \ldots, \boldsymbol{x}_{j,t-m+1}] \in \mathbb{R}^{n\times m}$, the feature-level null is a grouped null. Although this example is written as a lagged linear model, the same question appears whenever an original feature is represented by a structured block, including grouped regression and sequential networks, where it may affect output through multiple time steps, hidden states, or attention channels. The inferential target remains in feature-level, but the testing object is now its grouped input block.

Motivated by this grouped feature-level null, we propose a model-agnostic FDR control framework for grouped and sequential models. For linear lagged or grouped models, we develop a *Group Gaussian Mirror* procedure that replaces scalar mirror variables with block-level mirror constructions and yields null-symmetric group-level mirror statistics. For sequential neural networks, our method combines matrix-valued mirror perturbations with Derivatives of Permutation SHAP [38, 42], which provide model-agnostic block-level importance scores by differentiating each feature's SHAP value with respect to its sub-features. This makes the framework applicable to recurrent and attention-based architectures such as LSTM [29], GRU [12], and Transformer encoder [55] models.

We provide theoretical guarantees for both the linear and neural network versions of the framework. We prove null symmetry and FDR control for low- and high-dimensional grouped linear models, using a two-stage screen-then-mirror strategy in high dimensions. For neural models, we justify smoothed Permutation SHAP derivatives as asymptotically symmetric feature-importance scores under a fixed fitted nonlinear model and establish a symmetry-safe principle in high dimensions. Empirically, we evaluate the proposed framework across linear lagged models and sequential neural architectures, including LSTM, GRU, and Transformer encoder models, in both low- and high-dimensional settings. Our simulations cover various scenarios involving covariates correlation, lag-block dependence, nonlinear relationships and signal strength. Across simulated and real-world datasets, our method maintains reliable FDR control and improves power, especially when signals are distributed across correlated feature blocks. We provide detailed algorithms in Appendix A, flowchart in Appendix B and proofs of all theorems and propositions in Appendix C.

Our main contributions are threefold: we extend FDR control from coordinate-wise hypotheses to grouped-feature hypotheses; we provide a model-agnostic FDR framework for neural sequential models; and we introduce Permutation-SHAP derivatives as block-level feature importance scores for constructing the mirror-statistic in FDR control.

Notations: Uppercase letters denote matrices, bold lowercase letters denote vectors, and lowercase letters denote scalars. We write $X = [X_1, \ldots, X_p]$, where $X_j \in \mathbb{R}^{n\times m}$ is the block for feature j, and X_{-j} is the design with this block removed. Let P_A and $M_A = I_n - P_A$ denote projection and residual-maker matrices. Let $\mathcal{S}_1$ and $\mathcal{S}_0$ denote the sets of relevant and null features, respectively.

2 Background

2.1 Gaussian Mirror

Gaussian Mirror [60] controls FDR by constructing a pair of mirror variables $x_j^{\pm} = x_j \pm c_j z_j$ for each feature x_j, where $z_j \sim N(0, I_n)$ is independent Gaussian noise. The perturbation c_j is chosen so that x_j^+ and x_j^- are conditionally uncorrelated given X_{-j}. In a linear model, this makes the corresponding fitted weights $\widehat{\beta}_j^+$, $\widehat{\beta}_j^-$ independent centered Gaussian variables under the null $H_{0,j} : \beta_j = 0$, while relevant features tend to produce two mirror weights with aligned nonzero signal.

The key property for FDR control is the sign symmetry of the mirror statistic under null features. A best choice under most conditions is the "signed-sum" [13, 14]

$$M_j = \operatorname{sign}\left(\widehat{\beta}_j^+, \widehat{\beta}_j^-\right)\left(|\widehat{\beta}_j^+| + |\widehat{\beta}_j^-|\right).$$

For null features, M_j is symmetric around zero, whereas for relevant features it is likely to be large and positive. Hence the negative statistics provide an estimate of the number of false discoveries

among the positive statistics. For target FDR level q, it selects features with $M_j \geq \tau_q$, where [9]

$$\tau_q = \min\left\{t > 0 : \frac{\#\{j : M_j \leq -t\} + 1}{\#\{j : M_j \geq t\} \vee 1} \leq q\right\}.$$

This mirror statistic principle is the basis for our block-level extension.

2.2 Neural Gaussian Mirror

Neural Gaussian Mirror [59] extends the mirror construction to nonlinear models. In the linear case, the key construction target is conditional linear uncorrelation between the two mirror variables. For neural networks, this is replaced by the stronger conditional-independence target

$$U \perp V \mid W, \quad \text{where} \quad U = x_j + c_j z_j, \quad V = x_j - c_j z_j, \quad W = X_{-j}.$$

To obtain a computable objective, it places the log-density $\eta(u, v, w) = \log p_{U,V,W}(u, v, w)$ in a tensor-product Reproducing Kernel Hilbert Space (RKHS). Its interaction can be decomposed as

$$\eta = \eta_U + \eta_V + \eta_W + \eta_{U,W} + \eta_{V,W} + \eta_{U,V} + \eta_{U,V,W}.$$

The conditional-independence null is characterized by the absence of interaction components involving both mirror branches:

$$U \perp V \mid W \quad \Longleftrightarrow \quad \eta_{U,V} + \eta_{U,V,W} = 0.$$

More formally, each marginal RKHS is decomposed as $\mathcal{H}_A = \mathcal{H}_A^0 \oplus \mathcal{H}_A^1$ for $A \in \{U, V, W\}$, where $\mathcal{H}_A^0$ and $\mathcal{H}_A^1$ denote the constant and centered components. Thus the full tensor-product space admits the decomposition

$$\mathcal{H} = \bigoplus_{a,b,c \in \{0,1\}} \mathcal{H}_{abc} = \bigoplus_{a,b,c \in \{0,1\}} \mathcal{H}_U^a \otimes \mathcal{H}_V^b \otimes \mathcal{H}_W^c.$$

The conditional-independence null subspace and the alternative interaction subspace are

$$\mathcal{H}_0 = \mathcal{H}_{000} \oplus \mathcal{H}_{100} \oplus \mathcal{H}_{010} \oplus \mathcal{H}_{001} \oplus \mathcal{H}_{101} \oplus \mathcal{H}_{011}, \quad \mathcal{H}_1 = (\mathcal{H}_U^1 \otimes \mathcal{H}_V^1 \otimes \mathcal{H}_W^0) \oplus (\mathcal{H}_U^1 \otimes \mathcal{H}_V^1 \otimes \mathcal{H}_W^1).$$

The likelihood-ratio score in this alternative interaction space yields the kernel objective

$$I_j^K(c)^2 = \frac{1}{n^2}\left[(H_n K_U H_n) \circ (H_n K_V H_n) \circ K_W\right]_{++},$$

where $H_n = I_n - \mathbf{1}\mathbf{1}^\top/n$, $\circ$ is the Hadamard product, and $[]_{++}$ is the sum of all entries. Neural Gaussian Mirror chooses $\widehat{c}_j \in \arg\min_{c \geq 0} I_j^K(c)^2$. Our method for neural networks builds upon this RKHS conditional-dependence objective, while extending it to block-valued mirror variables.

3 Method for Linear Models

3.1 Block-Level Mirror Statistics

The original Gaussian Mirror is designed for coordinate-wise hypotheses, where one feature corresponds to one weight. In grouped models, the null object should be a whole weight block.

Proposition 1. *In a grouped design $X = [X_1, \ldots, X_p]$ with $X_j \in \mathbb{R}^{n \times m}$ and $\boldsymbol{\beta}_j \in \mathbb{R}^m$, applying the Gaussian Mirror separately to each sub-feature of X_j targets the coordinate-wise nulls $H_{0,(j,\ell)}$: $\beta_{j,\ell} = 0$ for $\ell = 0, \ldots, m-1$, rather than the grouped null $H_{0,j} : \boldsymbol{\beta}_j = (\beta_{j,0}, \ldots, \beta_{j,m-1}) = \mathbf{0}$.*

Proposition 1 shows that once the inferential target becomes grouped, the mirror construction must also be lifted from the scalar level to the block level. For each feature j, we therefore introduce a pair of block-level mirror variables of the form

$$X_j^+ = X_j + \widetilde{Z}_j G_j, \qquad X_j^- = X_j - \widetilde{Z}_j G_j.$$

where $\widetilde{Z}_j \in \mathbb{R}^{n \times m}$ is a transformed noise matrix and $G_j \in \mathbb{R}^{m \times m}$ is a group perturbation matrix. In the original Gaussian Mirror, the perturbation $c_j \boldsymbol{z}_j$ is chosen so that the two residualized mirror

variables are orthogonal. Here, the matrix perturbation $\widetilde{Z}_j G_j$ plays the same role at the group level, but now the relevant object is an $m \times m$ block cross-product rather than a scalar covariance term.

Let $\widehat{\boldsymbol{\beta}}_j^+$ and $\widehat{\boldsymbol{\beta}}_j^-$ denote the fitted weight vectors of X_j^+ and X_j^-, respectively, in the regression on $[X_{-j}, X_j^+, X_j^-]$. We then form a block-level mirror statistic from this pair to replace the scalar one, which preserves the properties of the previous "signed sum" in Section 2.1. A concrete example used throughout this paper is:

$$M_j = \operatorname{sign}\Big(\langle \widehat{\boldsymbol{\beta}}_j^+, \widehat{\boldsymbol{\beta}}_j^- \rangle\Big) \cdot \Big(\|\widehat{\boldsymbol{\beta}}_j^+\|_2 + \|\widehat{\boldsymbol{\beta}}_j^-\|_2\Big).$$

3.2 Low-Dimensional Group Gaussian Mirror

We now give the explicit Group Gaussian Mirror construction in low-dimensional settings. Fix a grouped feature j, let $R_j = M_{X_{-j}} X_j$ be the residualized group block, and $Z_j \in \mathbb{R}^{n \times m}$ be an i.i.d. Gaussian noise matrix independent of $(X, \boldsymbol{y})$. We define

$$Q_j = \big(I_n - P_{[X_{-j}, R_j]}\big) Z_j, \qquad \widetilde{Z}_j = Q_j \big(Q_j^\top Q_j\big)^{-1/2}, \qquad G_j = \big(R_j^\top R_j\big)^{1/2}.$$

The low-dimensional Group Gaussian Mirror variables are then constructed as $X_j^\pm = X_j \pm \widetilde{Z}_j G_j$.

Theorem 1 shows the grouped analogue exactly inherits the key properties of the original Gaussian Mirror. Given the block-level mirror statistics $\{M_j\}_{j=1}^p$, Group Gaussian Mirror uses the same mirror thresholding and selection rule as in 2.1, and effectively controls the FDR at the targeted level as shown in Proposition 2.

Theorem 1. *Assume that $R_j^\top R_j$ and $Q_j^\top Q_j$ are non-singular. For a grouped linear model with $X_j^\pm = X_j \pm \widetilde{Z}_j G_j$, the coefficient vectors $\widehat{\boldsymbol{\beta}}_j^+$ and $\widehat{\boldsymbol{\beta}}_j^-$ satisfy $\mathbb{E}[\widehat{\boldsymbol{\beta}}_j^+] = \mathbb{E}[\widehat{\boldsymbol{\beta}}_j^-] = \mathbf{0}$, $\operatorname{Cov}(\widehat{\boldsymbol{\beta}}_j^+) = \operatorname{Cov}(\widehat{\boldsymbol{\beta}}_j^-)$, $\operatorname{Cov}(\widehat{\boldsymbol{\beta}}_j^+, \widehat{\boldsymbol{\beta}}_j^-) = 0$, and are jointly Gaussian under $H_{0,j}$, which yields null symmetry of the block-level mirror statistic M_j. Alternatively, $\mathbb{E}[\widehat{\boldsymbol{\beta}}_j^+] = \mathbb{E}[\widehat{\boldsymbol{\beta}}_j^-] = \frac{1}{2}\boldsymbol{\beta}_j$ under $H_{1,j}$.*

Proposition 2. *Given the properties in Theorem 1, Group Gaussian Mirror asymptotically controls the FDR at the target level q.*

The following Proposition 3 also shows that the same construction can be extended to GLS estimators when we assume ε are not i.i.d.. In Experiment 5.1, however, we find the OLS version robust enough under correlated ε.

Proposition 3. *For a grouped linear model with gaussian noise $\varepsilon \sim N(0, \sigma^2\Sigma)$, where $\Sigma \succ 0$, let L satisfy $L^\top L = \Sigma^{-1}$, define $\bar{Y} = LY$ and $\bar{X}_k = LX_k$ for $k = 1, \ldots, p$. Then applying the OLS Group Gaussian Mirror to the whitened system $(\bar{Y}, \bar{X}, \bar{\varepsilon})$ is equivalent to applying GLS Group Gaussian Mirror to the original system (Y, X, ε) with Σ^{-1}-inner product.*

3.3 High-Dimensional Group Gaussian Mirror

When $pm > n$, the OLS-based Group Gaussian Mirror is no longer directly applicable. Moreover, the coordinate-wise post-selection argument used in the original high-dimensional Gaussian Mirror [60] relies on the polyhedral representation of the lasso selection event in the response vector. The following Proposition 4 shows that this argument is not invariant under the replacement from ℓ_1-Lasso to Group Lasso [62] and thus cannot be directly applied either.

Proposition 4. *For a grouped linear model, assume the weight obtained from the first-stage Group Lasso screener is $\widehat{\boldsymbol{\beta}}^{GL}$, and let $\widehat{\mathcal{A}} = \{j : \widehat{\boldsymbol{\beta}}_j^{GL} \neq \mathbf{0}\}$ be the selected grouped feature set. Then, in general, the event $\widehat{\mathcal{A}} = A$ is not polyhedral in Y.*

Therefore, we adopt a two-stage procedure in light of [13]. We split the sample into two independent (or asymptotically independent) chunks, perform first-stage screening on the first chunk, then use our algorithm in the second chunk. For time-series models, we divide the observations into two consecutive chunks and leave a short buffer gap between them; under stationarity and a suitable mixing condition, the cross-chunk dependence becomes asymptotically negligible [61, 8]. Let

$(X^{(1)}, \boldsymbol{y}^{(1)})$ and $(X^{(2)}, \boldsymbol{y}^{(2)})$ denote the two resulting data chunks. In the first stage, we apply a grouped feature screening method to the first chunk, and obtain a selected feature set $\widehat{\mathcal{A}} \subseteq \{1, \ldots, p\}$. In the second stage, we restrict the design to the selected grouped features $\{X_j^{(2)} : j \in \widehat{\mathcal{A}}\}$, and run exactly the same low-dimensional Group Gaussian Mirror procedure as in Section 3.2 on the second chunk only. For grouped features not selected in the first stage, their mirror statistics are set to zero.

Theorem 2. *Assume that: (i) the two data chunks share the same grouped-feature null structure and are independent, or asymptotically independent; (ii) the first-stage screening procedure satisfies a sure screening property, namely $\mathbb{P}(\mathcal{S}_1 \subseteq \widehat{\mathcal{A}}) \to 1$; and (iii) with probability tending to one, the second-stage restricted design is low-dimensional and full-rank. Then for every null grouped feature $j \in \mathcal{S}_0$, the two-stage mirror statistic $\widetilde{M}_j$ satisfies*

$$\sup_{t>0} \left| \mathbb{P}(\widetilde{M}_j > t) - \mathbb{P}(\widetilde{M}_j < -t) \right| \to 0.$$

Theorem 2 formalizes the key point that the first stage is used only for dimension reduction, while the null symmetry is inherited entirely from the second stage. Consequently, applying the standard mirror thresholding rule to $\{\widetilde{M}_j\}_{j=1}^p$ yields asymptotic FDR control. Under suitable conditions, Group Lasso is known to have consistent group-level sure screening [58] and therefore be our default choice. More generally, any method with the sure screening property can be used in the first stage.

4 Method for Neural Networks

4.1 Permutation SHAP Derivative as Feature Importance

For FDR control in neural networks, we first need a feature-importance metric that is stable and compatible with grouped-feature inference. Existing neural knockoff and Neural Gaussian Mirror methods typically use variants of fitted derivatives [28], obtained by perturbing one input coordinate while holding all others fixed. This is not well suited to our grouped or sequential setting. Perturbing an entire grouped feature jointly no longer yields a standard partial derivative, and differentiating each sub-feature separately does not directly represent the overall effect of the grouped feature and can be highly unstable when sub-features within a block are strongly correlated.

Therefore, we propose Permutation SHAP [38, 42] derivatives as our model-agnostic feature importance metric. For a fixed fitted nonlinear model f, let Π be a uniformly random permutation of the grouped feature indices, $P_j(\pi)$ denote the set of features appearing before feature j in a given permutation order π, X' be an independent background draw, and $(\boldsymbol{x}_S, \boldsymbol{X}'_{-S})$ denote the complete hybrid input vector whose feature blocks in S are taken from $\boldsymbol{x}$ and whose remaining blocks are taken from X'. The single-permutation marginal contribution of feature j is defined as

$$\Delta_j(\boldsymbol{x}; \pi, X') := f\big(\boldsymbol{x}_{P_j(\pi)\cup\{j\}}, X'_{-(P_j(\pi)\cup\{j\})}\big) - f\big(\boldsymbol{x}_{P_j(\pi)}, X'_{-P_j(\pi)}\big),$$

We then define the exact Permutation SHAP value $\Phi_j(\boldsymbol{x})$ and its derivative $\psi_j(\boldsymbol{x})$ by

$$\Phi_j(\boldsymbol{x}) := \mathbb{E}_{\Pi, X'}\left[\Delta_j(\boldsymbol{x}; \Pi, X')\right], \quad \text{and} \quad \boldsymbol{\psi}_j(\boldsymbol{x}) := \nabla_{\boldsymbol{x}_j} \Phi_j(\boldsymbol{x}).$$

In our grouped-feature setting, specifically, for grouped feature j with sub-features $x_{j,1}, \ldots, x_{j,m}$, we compute Permutation SHAP for the grouped feature and differentiate it with respect to each sub-feature, yielding a block-level importance vector analogous to $\boldsymbol{\beta}_j$ in linear models:

$$\boldsymbol{\psi}_j(\boldsymbol{x}) = \left(\frac{\partial \Phi_j(\boldsymbol{x})}{\partial x_{j,1}}, \ldots, \frac{\partial \Phi_j(\boldsymbol{x})}{\partial x_{j,m}} \right)^\top.$$

For each sub-feature $x_{j,i}$, this method produces n derivative values across the n explained points. We aggregate them by taking the average: $\widehat{\boldsymbol{\psi}}_j^{\pm} = \frac{1}{n}\sum_{t=1}^n \widehat{\boldsymbol{\psi}}_j^{\pm}(\boldsymbol{x}_t)$. This aggregation targets the average SHAP-derivative effect; hence the neural grouped-feature null is interpreted as zero average block-level effect after aggregation, without requiring the local derivative effects to be time-invariant. The neural mirror statistic M_j is defined similarly to the construction in Section 3.1:

$$M_j = \operatorname{sign}\left(\langle \widehat{\boldsymbol{\psi}}_j^{+}, \widehat{\boldsymbol{\psi}}_j^{-} \rangle\right) \left(\|\widehat{\boldsymbol{\psi}}_j^{+}\|_2 + \|\widehat{\boldsymbol{\psi}}_j^{-}\|_2 \right).$$

Proposition 5. *For a grouped linear model, let $\Phi_j(\boldsymbol{x})$ be the group-level Permutation SHAP value of group j. Then $\Phi_j(\boldsymbol{x}) = \boldsymbol{\beta}_j^\top \left(\boldsymbol{x}_j - \mathbb{E}[X'_j]\right)$ and hence $\boldsymbol{\psi}_j(\boldsymbol{x}) = \nabla_{\boldsymbol{x}_j}\Phi_j(\boldsymbol{x}) = \boldsymbol{\beta}_j$, which coincides with the linear-case coefficient vector $\boldsymbol{\beta}_j$.*

These Permutation SHAP derivatives can be viewed as a coalition-weighted fitted derivatives, and Proposition 5 shows that it is a direct nonlinear analogue of the coefficient vector in a grouped linear model. In practice, we approximate $\Phi_j(\boldsymbol{x})$ by Monte Carlo Permutation SHAP and estimate its derivative by local-quadratic smoothing with a tricube kernel and $\lceil 0.3n \rceil$-nearest-neighbor bandwidth.

Theorem 3. *For a fixed nonlinear model f and a null grouped feature j, let $\widehat{\psi}_{j,k}(x_0)$ denote the local-quadratic smoothed Monte Carlo Permutation SHAP derivative estimator. Under some regularity conditions, we have, $\widehat{\psi}_{j,k}(x_0)$ is symmetric around zero for each sub-feature $x_{j,k}$, namely,*

$$\sup_{t>0}\left|\mathbb{P}\Big(\widehat{\psi}_{j,k}(x_0) > t\Big) - \mathbb{P}\Big(\widehat{\psi}_{j,k}(x_0) < -t\Big)\right| \to 0, \quad \forall k.$$

Theorem 3 justifies using the aggregated block-level SHAP derivative vector as the neural mirror importance score. In the experiment, we find that the number of permutation draws does not affect the result heavily. As detailed in Appendix E, we find that 10 draws are sufficient and performs stably, and we fix this number in our later experiments.

4.2 Low-Dimensional Permutation SHAP Gaussian Mirror

We now introduce our method in neural networks, which combines the block-level Permutation SHAP feature importance in Section 4.1 with a grouped Neural Gaussian Mirror construction. The mirror variables follow the linear Group Gaussian Mirror. For feature j, we define

$$U_j(G_j) = X_j + \widetilde{Z}_j G_j, \quad V_j(G_j) = X_j - \widetilde{Z}_j G_j, \quad W_j = X_{-j}$$

where $\widetilde{Z}_j \in \mathbb{R}^{n\times m}$ is the Gaussian noise block and $G_j \in \mathbb{R}^{m\times m}$ is the perturbation matrix. The difference is that G_j can no longer be chosen by the closed-form block orthogonality condition in Section 3.2; instead, it is optimized through the kernel-based conditional-dependence objective introduced in Section 2.2. The target remains $U_j(G_j) \perp V_j(G_j) \mid W_j$, while $U_j(G_j)$ and $V_j(G_j)$ are now being block-valued mirror branches.

Theorem 4. *Applying RKHS log-density decomposition and likelihood-ratio score construction to $(U_j(G_j), V_j(G_j), W_j)$ gives the grouped Neural Gaussian Mirror objective:*

$$I_j^K(G_j)^2 = \frac{1}{n^2}\left[(H_n K_U(G_j) H_n) \circ (H_n K_V(G_j) H_n) \circ K_W\right]_{++},$$

where K_U, K_V and K_W are kernel matrices computed from the rows of $U_j(G_j)$, $V_j(G_j)$ and W_j.

Theorem 4 shows that the grouped construction changes the perturbation G_j from a scalar to a matrix, while the conditional dependence criterion remains a scalar RKHS score norm. In all experiments, K_U, K_V, K_W are Gaussian RBF kernels with median pairwise-distance bandwidths after column standardization. We therefore choose $\widehat{G}_j \in \arg\min_{G_j\in\mathcal{G}} I_j^K(G_j)^2$.

Proposition 6. *Under linear kernels and ignoring the W_j-kernel weight, the objective satisfies*

$$I_j^K(G_j)^2 \propto \left\| \frac{1}{n}\big(H_n U_j(G_j)\big)^\top \big(H_n V_j(G_j)\big) \right\|_F^2,$$

which reduces to the block-orthogonality target in the linear Group Gaussian Mirror.

Proposition 6 shows that the grouped neural construction reduces to the block-orthogonality principle of linear GGM under linear kernels. Empirically, the objective $I_j^K(G_j)^2$ is well behaved: full-matrix optimization initialized at I_m converges stably even with a relatively large learning rate. Appendix F provides the convergence results, and we fix learning rate 0.5 and 200 epochs in later experiments.

4.3 High-Dimensional Permutation SHAP Gaussian Mirror

For high-dimensional neural networks, the data-splitting and Group Lasso screening in Section 3.3 are not directly applicable, since they are designed for OLS or GLS inference; exact OLS residualization

from Section 3.2 is also unavailable. We therefore use a label-free ridge-type projection to construct a stable perturbation basis $\widetilde{Z}_{j,\lambda}$. This projection is not meant to remove nonlinear dependence, which is handled by $I_j^K(G_j)^2$, but to place mirror noise in a stable incremental direction while preserving the sign-flip symmetry required by the mirror principle. We define

$$Q_{j,\lambda} = A_{j,\lambda}(X_j, W_j)Z_j, \quad \widetilde{Z}_{j,\lambda} = Q_{j,\lambda}\left(Q_{j,\lambda}^\top Q_{j,\lambda} + \varepsilon I_m\right)^{-1/2}, \quad \text{with} \quad \varepsilon > 0.$$

Here, $A_{j,\lambda}(X_j, W_j) \in \mathbb{R}^{n\times n}$ is a ridge-type projection operator, and $Z_j \in \mathbb{R}^{n\times m}$ is an auxiliary Gaussian noise matrix independent of $(X, \boldsymbol{y})$ and $A_{j,\lambda}(X_j, W_j)$. In later high-dimensional experiments, $A_{j,\lambda}$ is implemented by ridge residualization using ridge hat matrices, with $\lambda_W = \lambda_Z = 0.1n$.

Theorem 5. *For any grouped feature j, let $Z_j \in \mathbb{R}^{n\times m}$ independently generated and satisfy $Z_j \overset{d}{=} -Z_j$ and $Z_j \perp (X, \boldsymbol{y})$. Assume $\widetilde{Z}_{j,\lambda}$ is odd and $\widehat{G}_j$ is even in Z_j, and the downstream training-and-attribution procedure is swap-equivariant in distribution under exchanging the two mirror branches. Then the mirror outputs $O_j^\pm$ are exchangeable in distribution, namely, $(O_j^+, O_j^-) \overset{d}{=} (O_j^-, O_j^+)$.*

Theorem 5 shows that the validity does not rely on exact OLS orthogonality. This theorem gives a structural requirement that the construction pipeline should be symmetry-safe. The ridge-type operator $A_{j,\lambda}$ satisfies the requirement that the projection is odd in Z_j, as it is label-free and linear in Z_j. Thus, the ridge-type projection preserves all the geometry required by the mirror principle. Combined with the Permutation SHAP derivative metric from Section 4.1, which is swap-equivariant, our high-dimensional Neural Gaussian Mirror construction preserves the required structural exchangeability. Together, Theorems 3 and 5 establish key theoretical properties underlying the validity of PSGM under their respective assumptions. However, extending these guarantees to a full FDR control guarantee for the trained neural network requires additional conditions on systematic null-feature attribution. This challenge is closely related to predictive and explanation multiplicity and the Rashomon set, as further discussed in Appendix D.

5 Numerical Simulation

5.1 Linear Model Simulations

We evaluate our Group Gaussian Mirror (GGM) method on grouped linear designs, using both the OLS and the GLS version and compare it with simple Group Lasso. We also compare it with a Partial-F BH procedure, where we compute group-level p-values by partial F-tests for $H_{0,j} : \boldsymbol{\beta}_j = \mathbf{0}$, and apply the standard Benjamini–Hochberg procedure to the resulting p-values [6]. In all settings, data are generated from $\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where the noise sequence $\{\varepsilon_t\}_{t=1}^n$ follows a stationary AR(1) process $\varepsilon_t = \phi\varepsilon_{t-1} + u_t$, $u_t \overset{\text{iid}}{\sim} N(0, 1-\phi^2)$, $\varepsilon_0 \sim N(0,1)$ with $\phi = 0.5$. We set grouped feature size $m = 5$, target FDR level $q = 0.1$, and $|\mathcal{S}_1| = 30$ relevant grouped features. All weights in null grouped features are set to zero. For relevant grouped features, the signal scale is $\sigma_\beta = 20\sqrt{\log(|\mathcal{S}_1|)/n}$. We use an AR grouped design: for each non-null grouped feature j, we draw $a_j \sim N(0, \sigma_\beta^2)$, then set $\beta_{j,\ell} = w_\ell a_j$, where w_ℓ decreases linearly from 1.0 to 0.4 across lags. This creates signals that are spread across the whole lag block but weaker at more distant lags. We consider two dimension settings: a low-dimensional setting with $n = 2500$, $p = 300$, $m = 5$, and a high-dimensional setting with $n = 2500$, $p = 700$, $m = 5$. In the high-dimensional setting, we use the screen-then-mirror version of Group Gaussian Mirror, splitting the observations into two consecutive blocks with a gap of 10 observations between them. All experiments were carried out on a Supermicro AS-4124GO-NART+ server with eight NVIDIA A100-SXM4-80GB GPUs.

Each grouped feature is generated from m lagged values of a latent stationary AR(1) process. We vary the feature-level AR coefficient $\rho_w \in \{0.1, 0.5, 0.9\}$, which controls the temporal dependence among lagged sub-features within the same grouped feature. Cross-feature dependence is introduced through the contemporaneous correlation $\rho_b = \alpha\rho_w$ between latent processes, with $\alpha \in \{0.2, 0.8\}$.

For each setting, we repeat the experiment 50 times and report grouped-feature FDR and power averaged over repetitions. FDR is computed as the fraction of selected grouped features that are null, and power is computed as the fraction of non-null grouped features selected. The results are summarized in Table 1. In all tables below, **bold** numbers represent the best result and blue numbers represent the second best. Empirically, we find the OLS version is sufficient even for correlated noise

terms. We also conduct additional linear model experiments in Appendix H, including experiments under a block-correlated Gaussian design and comparison with the Group Knockoff baseline [15].

Table 1: Linear simulation results under the AR grouped design

Setting (p*m, n)	Feature Corr	AR Coef	GGM-OLS		GGM-GLS		Partial-F BH		Group Lasso	
			FDR	Power	FDR	Power	FDR	Power	FDR	Power
$p*m=1500$ $n=2500$	0.2	0.1	0.135	1.000	0.119	1.000	**0.102**	1.000	0.412	1.000
		0.5	**0.020**	1.000	0.069	1.000	0.053	1.000	0.402	1.000
		0.9	**0.026**	1.000	0.090	1.000	0.069	1.000	0.457	0.999
	0.8	0.1	0.077	1.000	0.078	1.000	**0.062**	1.000	0.367	1.000
		0.5	**0.043**	0.997	0.129	0.997	0.065	0.997	0.402	0.999
		0.9	**0.024**	0.995	0.090	0.995	0.081	0.995	0.449	0.996
$p*m=3500$ $n=2500$	0.2	0.1	0.104	1.000	0.114	1.000	**0.080**	1.000	0.402	1.000
		0.5	**0.033**	1.000	0.067	1.000	0.063	1.000	0.408	1.000
		0.9	**0.024**	1.000	0.081	1.000	0.076	0.998	0.463	1.000
	0.8	0.1	0.087	1.000	0.092	0.996	**0.058**	1.000	0.389	0.997
		0.5	**0.046**	1.000	0.123	1.000	0.062	0.997	0.427	1.000
		0.9	**0.019**	0.997	0.099	0.998	0.085	1.000	0.455	0.992

5.2 Sequential Neural Network Simulations

Table 2: LSTM simulation results

Setting (p*m, n)	Signal Strength	PSGM		Fitted Derivative		AGL		AFS		Sawaya	
		FDR	Power	FDR	Power	FDR	Power	FDR	Power	FDR	Power
$p*m=1500$ $n=2500$	Weak	**0.161**	0.879	0.213	0.842	0.916	**0.966**	0.886	0.556	0.448	0.007
	Strong	**0.145**	**0.958**	0.265	0.790	0.833	0.880	0.644	0.723	0.276	0.029
$p*m=3500$ $n=2500$	Weak	**0.116**	0.625	0.146	0.649	0.938	**0.960**	0.968	0.311	0.493	0.007
	Strong	**0.152**	**0.886**	0.188	0.763	0.854	0.880	0.912	0.683	0.519	0.019

Table 3: GRU simulation results

Setting (p*m, n)	Signal Strength	PSGM		Fitted Derivative		AGL		AFS		Sawaya	
		FDR	Power	FDR	Power	FDR	Power	FDR	Power	FDR	Power
$p*m=1500$ $n=2500$	Weak	**0.133**	0.931	0.239	0.821	0.837	0.899	0.555	**0.944**	0.476	0.176
	Strong	**0.162**	0.979	0.249	0.854	0.764	0.840	0.220	**0.992**	0.600	0.203
$p*m=3500$ $n=2500$	Weak	**0.062**	0.817	0.134	0.851	0.947	**1.000**	0.590	0.821	0.577	0.024
	Strong	**0.223**	0.919	0.308	0.889	0.829	0.880	0.441	**1.000**	0.478	0.163

Table 4: Transformer simulation results

Setting (p*m, n)	Signal Strength	PSGM		Fitted Derivative		AGL		AFS		Sawaya	
		FDR	Power	FDR	Power	FDR	Power	FDR	Power	FDR	Power
$p*m=1500$ $n=2500$	Weak	**0.068**	0.846	0.098	0.855	0.948	**1.000**	0.668	0.591	0.224	0.906
	Strong	**0.116**	0.887	0.232	**0.893**	0.814	0.880	0.347	0.872	0.306	0.683
$p*m=3500$ $n=2500$	Weak	**0.141**	0.491	0.183	0.577	0.974	**1.000**	0.739	0.475	0.278	0.667
	Strong	0.283	0.900	**0.160**	0.710	0.903	**0.925**	0.720	0.843	0.432	0.619

We evaluate our Permutation SHAP Gaussian Mirror (PSGM) method on nonlinear sequential designs using LSTM, GRU, and Transformer encoder models. The target FDR level is still $q = 0.1$, and we fix $|\mathcal{S}_1| = 15$ to mimic a sparse setting. As in linear simulations, each grouped feature has lag-block length consistent with the model's look-back window m. We also consider a low-dimensional setting with $n = 2500$, $p = 300$, $m = 5$, and a high-dimensional setting with $n = 2500$, $p = 700$, $m = 5$. We also conduct runtime/memory complexity and performance sensitivity analysis of PSGM to different scales of p, n, m in both low- and high-dimensional settings in Appendix G.

The covariates are generated from a latent factor design similar to [66]. At each time t, we sample a 10-dimensional latent factor vector with correlated coordinates and smooth it over time:

$$\boldsymbol{f}_t^{\text{raw}} \sim N(\mathbf{0}, \Sigma_f), \quad (\Sigma_f)_{ab} = 0.9^{|a-b|}, \quad \boldsymbol{f}_t = 0.3\boldsymbol{f}_{t-1}^{\text{raw}} + 0.7\boldsymbol{f}_t^{\text{raw}}.$$

Given $\boldsymbol{f}_t$, all features are generated by the same logistic factor mechanism. For each feature j, we independently draw $c_j \sim N(0,1)$ and $\tilde{\boldsymbol{\lambda}}_j \sim N(\mathbf{0}, I_{11})$. Let $\tilde{\boldsymbol{f}}_t = (1, \boldsymbol{f}_t^\top)^\top \in \mathbb{R}^{11}$. We set

$$X_{tj} = \frac{c_j}{1 + \exp(\tilde{\boldsymbol{\lambda}}_j^\top \tilde{\boldsymbol{f}}_t)} + e_{tj}, \quad \text{where} \quad e_{tj} \sim N(0,1).$$

The relevant feature set is then sampled uniformly at random; all features are generated by the same mechanism, and relevance only determines whether a feature enters the response. For each relevant feature j, we assign a signed signal weight $\beta_j = 10 s_j$, where $s_j \in \{-1, 1\}$, while null features have $\beta_j = 0$. The response is generated from a lag-weighted nonlinear signal,

$$\eta_t = \sum_{\ell=0}^{m-1} w_\ell \, \boldsymbol{x}_{t-1-\ell}^\top \boldsymbol{\beta}, \qquad y_t = \tanh(\eta_t) + \varepsilon_t, \qquad \varepsilon_t \sim N(0, \sigma_\varepsilon^2),$$

where $w_\ell \propto 0.9^\ell$ is normalized to have unit ℓ_2-norm. We consider $\sigma_\varepsilon \in \{1, 0.3\}$, corresponding to weak and strong signal-to-noise regimes.

For each dataset, we train LSTM, GRU, and Transformer encoder models. The recurrent models use two layers, the Transformer uses two attention heads, and the hidden dimension is set to $20\log(p)$. All models are trained with MSE loss using Adam, with learning rate, weight decay, and dropout selected by the same cross-validation procedure. We compare our algorithm using Permutation SHAP derivative with ablations using the fitted derivative at sub-feature for importance score. We also compare with three baselines: AFS [26], Adaptive Group Lasso (AGL) [17], and Sawaya's data-splitting FDR-control variant [48]. The first two are feature-selection heuristics and do not guarantee FDR control, while the third is designed for controlled neural feature selection. For each model, dimension setting, and noise level, we repeat the experiment 50 times, and report the averaged grouped-feature FDR and power. The results are summarized in Table 2, Table 3, and Table 4.

6 Real Data Application: CausalRivers

We evaluate our method on the East Germany subset of CausalRivers [51] dataset, a real-world river-discharge time-series benchmark with 666 gauge stations and a directed upstream-to-downstream river graph. We aggregate the original 15-minute measurements to 3-hour averages, interpolate missing values, and use first differences to reduce persistent level effects. For each target station j, the real response is $y_t^{\text{real}} = \Delta R_{t,j}$, and each candidate station k forms one grouped lag feature $X_{t,k} = (\Delta R_{t-1,k}, \Delta R_{t-2,k}, \Delta R_{t-3,k})$. Thus, selection is performed at the station-block level rather than the individual-lag level. We fix the target FDR level $q = 0.2$ for a less conservative selection.

Table 5: LSTM results on the CausalRivers real data application

Outcome	Method	FDR/Proxy	Power/Proxy	% Direct	% 2nd	% Null	Sel. MSE	Full MSE	Oracle MSE
Real y	PSGM	0.739	0.299	0.331	0.276	0.212	**0.385**	0.436	0.365
	AGL	0.772	**1.000**	1.000	1.000	1.000	0.427	0.436	0.365
	AFS	0.795	0.140	0.131	0.150	0.155	0.598	0.436	0.365
	Sawaya	**0.405**	0.029	0.036	0.017	0.020	0.613	0.436	0.365
Semi-synthetic y	PSGM	**0.306**	0.896	0.992	0.826	0.143	**0.512**	0.575	0.501
	AGL	0.772	**1.000**	1.000	1.000	1.000	0.592	0.575	0.501
	AFS	0.919	0.056	0.064	0.059	0.178	0.865	0.575	0.501
	Sawaya	0.342	0.040	0.044	0.025	0.023	0.837	0.575	0.501

Table 6: GRU results on the CausalRivers real data application

Outcome	Method	FDR/Proxy	Power/Proxy	% Direct	% 2nd	% Null	Sel. MSE	Full MSE	Oracle MSE
Real y	PSGM	0.700	0.379	0.403	0.367	0.245	**0.373**	0.439	0.378
	AGL	0.772	**1.000**	1.000	1.000	1.000	0.437	0.439	0.378
	AFS	0.757	0.162	0.172	0.156	0.142	0.650	0.439	0.378
	Sawaya	**0.458**	0.057	0.051	0.062	0.022	0.513	0.439	0.378
Semi-synthetic y	PSGM	**0.264**	0.902	0.983	0.854	0.110	**0.508**	0.592	0.523
	AGL	0.772	**1.000**	1.000	1.000	1.000	0.587	0.592	0.523
	AFS	0.916	0.068	0.111	0.045	0.198	0.841	0.592	0.523
	Sawaya	0.347	0.052	0.078	0.043	0.017	0.819	0.592	0.523

Table 7: Transformer results on the CausalRivers real data application

Outcome	Method	FDR/Proxy	Power/Proxy	% Direct	% 2nd	% Null	Sel. MSE	Full MSE	Oracle MSE
Real y	PSGM	0.576	0.406	0.431	0.407	0.155	0.333	0.399	0.345
	AGL	0.772	**1.000**	1.000	1.000	1.000	0.407	0.399	0.345
	AFS	0.616	0.219	0.350	0.132	0.122	**0.332**	0.399	0.345
	Sawaya	**0.307**	0.092	0.151	0.048	0.028	0.382	0.399	0.345
Semi-synthetic y	PSGM	**0.258**	0.897	0.992	0.825	0.115	**0.514**	0.586	0.509
	AGL	0.772	**1.000**	1.000	1.000	1.000	0.597	0.586	0.509
	AFS	0.320	0.452	0.763	0.249	0.063	0.593	0.586	0.509
	Sawaya	0.340	0.124	0.233	0.035	0.023	0.720	0.586	0.509

We use the first 2400 differenced observations after January 1, 2021 with a train-test ratio of $4:1$. We consider six target stations selected by a fixed rule requiring at least 10 graph-relevant groups, 50 eligible graph-null groups, low-dimensional OLS feasibility, and acceptable condition numbers. Direct upstream and upstream-of-upstream stations are treated as graph-proxy relevant groups, while graph-null groups are sampled from stations with no directed path relation to the target. Since the river graph does not define the statistical null in our regression model, FDR and power on real y are reported only as graph-proxy metrics. To evaluate selection with known support under real covariate dependence, we also generate semi-synthetic responses on the same river covariates, using direct and second-order upstream groups as the true support. For each relevant group k, let $s_{t,k} = \boldsymbol{w}^\top X_{t,k}$, where $\boldsymbol{w}$ is an ℓ_2-normalized exponentially decaying lag-weight vector with larger weight on recent lags. We generate

$$\eta_t = \sum_{k \in \mathcal{S}_{\text{direct}}} \xi_k \tanh(s_{t,k}) + 0.55 \sum_{k \in \mathcal{S}_{\text{second}}} \xi_k \sin(s_{t,k}) + 0.20 \sum_{(k,k') \in \mathcal{P}_{\text{direct}}} \tanh(s_{t,k} s_{t,k'}/2),$$

where $\xi_k \in \{-1, 1\}$ and $\mathcal{P}_{\text{direct}}$ contains pairs of direct upstream groups. We set $y_t^{\text{syn}} = \eta_t + \varepsilon_t$, with Gaussian noise calibrated to signal-to-noise ratio 2.0. For semi-synthetic y, FDR and power are computed using the constructed true support. Table 5, 6, and 7 reports LSTM, GRU and Transformer results averaged over six target stations and ten random draws of graph-null groups, including FDR/proxy FDR, power/proxy power, selected-group composition, and held-out test MSE for the selected, full, and oracle upstream-only models. Full dataset description, preprocessing details, target-level results, and linear model results are given in Appendix I.

7 Conclusion and Limitations

We propose a model-agnostic FDR-control framework for grouped-feature selection in sequential models. To the best of our knowledge, this is the first Gaussian-Mirror-based method that extends mirror-statistic FDR control to sequential deep learning architectures while directly targeting grouped-feature null hypotheses. We introduce GGM for linear models and PSGM for neural networks. Across simulations and the CausalRivers application, our method achieves stronger FDR-power tradeoffs than benchmark methods while maintaining competitive predictive performance.

There are several limitations of our method. The neural version depends on the quality of the fitted predictive model, or how the fitted predictive model uses its input features. Null symmetry of PSGM mirror statistics may be weakened if the network learns spurious correlations or assigns systematic directional importance to null grouped features. This issue is closely related to the broader challenges of predictive and explanation multiplicity [16, 20, 21, 35, 22]. We discuss this issue further in Appendix D and identify it as an important direction for future research. In our experiments, stronger regularization, particularly greater weight decay, generally improves FDR control. PSGM also performs best when the grouped-feature size m is moderate and the relevant support is sparse; larger lag windows or denser signals may weaken mirror-statistic separation. Finally, our current method is relatively power-oriented and can sometimes be less conservative in FDR control, so designing more conservative neural mirror procedures is another useful future direction.

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Appendix

A Algorithms for GGM and PSGM

Algorithm 1 Low-Dimensional Group Gaussian Mirror

Input: Fixed FDR level q, grouped design $X = [X_1, \ldots, X_p]$ with $X_j \in \mathbb{R}^{n\times m}$, and $\boldsymbol{y} \in \mathbb{R}^n$.
Output: Selected grouped feature set $\widehat{\mathcal{S}}_1$.

(a) **for** $j = 1, \ldots, p$, **do**:

(1) Let X_{-j} be the design matrix with group j removed, and compute $R_j = M_{X_{-j}} X_j$.

(2) Generate an auxiliary Gaussian noise matrix $Z_j \in \mathbb{R}^{n\times m}$, with $Z_j \perp (X, \boldsymbol{y})$.

(3) Orthogonalize and normalize the noise block to get Q_j and $\widetilde{Z}_j$.

(4) Set the group perturbation matrix $G_j = (R_j^\top R_j)^{1/2}$.

(5) Construct the block-level mirror variables $X_j^+ = X_j + \widetilde{Z}_j G_j$ and $X_j^- = X_j - \widetilde{Z}_j G_j$.

(6) Fit the linear regression of $\boldsymbol{y}$ on $[X_{-j}, X_j^+, X_j^-]$ and obtain $\widehat{\boldsymbol{\beta}}_j^{\pm}$.

(7) Compute the block-level mirror statistic $M_j = \operatorname{sign}\left(\langle \widehat{\boldsymbol{\beta}}_j^+, \widehat{\boldsymbol{\beta}}_j^- \rangle\right)\left(\|\widehat{\boldsymbol{\beta}}_j^+\|_2 + \|\widehat{\boldsymbol{\beta}}_j^-\|_2\right)$.

(b) **end for**

(c) Compute the mirror threshold $\tau_q = \min\left\{t > 0 : \frac{\#\{j : M_j \le -t\}+1}{\#\{j : M_j \ge t\} \vee 1} \le q\right\}$.

(d) Return $\widehat{\mathcal{S}}_1 = \{j : M_j \ge \tau_q\}$.

Algorithm 2 High-Dimensional Two-Stage Group Gaussian Mirror

Input: Fixed FDR level q, grouped design $X = [X_1, \ldots, X_p]$ with $X_j \in \mathbb{R}^{n\times m}$, response $\boldsymbol{y} \in \mathbb{R}^n$, and first-stage grouped screener.
Output: Selected grouped feature set $\widehat{\mathcal{S}}_1$.

(a) Split the data into two chunks $(X^{(1)}, \boldsymbol{y}^{(1)})$ and $(X^{(2)}, \boldsymbol{y}^{(2)})$. For time-series data, use two consecutive chunks and leave a short buffer gap if needed.

(b) Apply a grouped feature screening method (Group Lasso) on the first chunk and obtain $\widehat{\mathcal{A}} \subseteq \{1, \ldots, p\}$, and initialize $\widetilde{M}_j = 0$, for $j = 1, \ldots, p$.

(c) **for** $j \in \widehat{\mathcal{A}}$, **do**:

(1) Restrict the second-stage design to the screened model $X^{(2)}_{\widehat{\mathcal{A}}} = \{X_k^{(2)} : k \in \widehat{\mathcal{A}}\}$.

(2) Compute the residualized group block: $R_j^{(2)} = M_{X^{(2)}_{\widehat{\mathcal{A}}\setminus\{j\}}} X_j^{(2)}$.

(3) Generate a Gaussian noise matrix $Z_j^{(2)} \in \mathbb{R}^{n_2\times m}$, with $Z_j^{(2)} \perp (X^{(2)}, \boldsymbol{y}^{(2)})$.

(4) Orthogonalize and normalize the noise block to get $Q_j^{(2)}$, $\widetilde{Z}_j^{(2)}$ and $G_j^{(2)}$.

(5) Construct second-stage mirror variables $X_j^{\pm,(2)} = X_j^{(2)} \pm \widetilde{Z}_j^{(2)} G_j^{(2)}$.

(6) Regress $\boldsymbol{y}^{(2)}$ on $[X^{(2)}_{\widehat{\mathcal{A}}\setminus\{j\}}, X_j^{+,(2)}, X_j^{-,(2)}]$ and obtain $\widehat{\boldsymbol{\beta}}_j^{\pm,(2)}$.

(7) Compute $\widetilde{M}_j = M_j^{(2)} = \operatorname{sign}\left(\langle \widehat{\boldsymbol{\beta}}_j^{+,(2)}, \widehat{\boldsymbol{\beta}}_j^{-,(2)} \rangle\right)\left(\|\widehat{\boldsymbol{\beta}}_j^{+,(2)}\|_2 + \|\widehat{\boldsymbol{\beta}}_j^{-,(2)}\|_2\right)$.

(d) **end for**

(e) Compute the mirror threshold $\tau_q = \min\left\{t > 0 : \frac{\#\{j : \widetilde{M}_j \le -t\}+1}{\#\{j : \widetilde{M}_j \ge t\} \vee 1} \le q\right\}$.

(f) Return $\widehat{\mathcal{S}}_1 = \{j : \widetilde{M}_j \ge \tau_q\}$.

Algorithm 3 Low-Dimensional Permutation SHAP Gaussian Mirror

Input: Fixed FDR level q, grouped design $X = [X_1, \ldots, X_p]$ with $X_j \in \mathbb{R}^{n\times m}$, response $\boldsymbol{y} \in \mathbb{R}^n$, model class $\mathcal{F}$, kernels k_U, k_V, k_W, permutation draws M_n, smoothing bandwidth h_n.
Output: Selected grouped feature set $\widehat{\mathcal{S}}_1$.

(a) **for** $j = 1, \ldots, p$, **do**:

(1) Construct an orthogonalized Gaussian noise block $\widetilde{Z}_j \in \mathbb{R}^{n\times m}$, with $\widetilde{Z}_j \perp (X, \boldsymbol{y})$.

(2) For each candidate $G_j \in \mathcal{G}$, construct $U_j(G_j), V_j(G_j) = X_j \pm \widetilde{Z}_j G_j$.

(3) Optimize the perturbation matrix and get $\widehat{G}_j \in \arg\min_{G_j\in\mathcal{G}} I_j^K(G_j)^2$.

(4) Construct the final neural mirror branches $U_j = X_j + \widetilde{Z}_j\widehat{G}_j$ and $V_j = X_j - \widetilde{Z}_j\widehat{G}_j$.

(b) **end for**

(c) Train a neural model $\widehat{f} \in \mathcal{F}$ using the augmented input $[U_1, V_1, \ldots, U_p, V_p]$ and $\boldsymbol{y}$.

(d) **for** $j = 1, \ldots, p$, **do**:

(1) For each sub-feature k, estimate the local-quadratic smoothed Permutation SHAP derivatives $\widehat{\psi}^{\pm}_{j,k}(\boldsymbol{x}_t)$ for each t. Aggregate them and get $\widehat{\boldsymbol{\psi}}_j^{\pm} = \frac{1}{n}\sum_{t=1}^n \widehat{\boldsymbol{\psi}}_j^{\pm}(\boldsymbol{x}_t)$.

(2) Compute the neural mirror statistic $M_j = \operatorname{sign}\Big(\langle \widehat{\boldsymbol{\psi}}_j^{+}, \widehat{\boldsymbol{\psi}}_j^{-}\rangle\Big)\Big(\|\widehat{\boldsymbol{\psi}}_j^{+}\|_2 + \|\widehat{\boldsymbol{\psi}}_j^{-}\|_2\Big)$.

(e) **end for**

(f) Compute the mirror threshold $\tau_q = \min\Big\{t > 0 : \frac{\#\{j:M_j\le -t\}+1}{\#\{j:M_j\ge t\}\vee 1} \le q\Big\}$.

(g) Return $\widehat{\mathcal{S}}_1 = \{j : M_j \ge \tau_q\}$.

Algorithm 4 High-Dimensional Ridge-Projected Permutation SHAP Gaussian Mirror

Input: Fixed FDR level q, grouped design $X = [X_1, \ldots, X_p]$ with $X_j \in \mathbb{R}^{n\times m}$, response $\boldsymbol{y} \in \mathbb{R}^n$, model class $\mathcal{F}$, kernels k_U, k_V, k_W, permutation draws M_n, smoothing bandwidth h_n.
Output: Selected grouped feature set $\widehat{\mathcal{S}}_1$.

(a) **for** $j = 1, \ldots, p$, **do**:

(1) Generate Gaussian noise matrix $Z_j \in \mathbb{R}^{n\times m}$, with $Z_j \overset{d}{=} -Z_j$ and $Z_j \perp (X, \boldsymbol{y})$.

(2) Construct a label-free ridge-type projection operator $A_{j,\lambda}(X_j, W_j) \in \mathbb{R}^{n\times n}$.

(3) Form the ridge-projected noise block $Q_{j,\lambda}$, normalize it and get $\widetilde{Z}_{j,\lambda}$.

(4) For each candidate $G_j \in \mathcal{G}$, construct $U_j(G_j), V_j(G_j) = X_j \pm \widetilde{Z}_{j,\lambda} G_j$.

(5) Initialize the perturbation matrix $G_j^{(0)} = I_m$, optimize $\widehat{G}_j \in \arg\min_{G_j\in\mathcal{G}} I_j^K(G_j)^2$.

(6) Construct the final mirror branches $U_j = X_j + \widetilde{Z}_{j,\lambda}\widehat{G}_j$ and $V_j = X_j - \widetilde{Z}_{j,\lambda}\widehat{G}_j$.

(b) **end for**

(c) Train a neural model $\widehat{f} \in \mathcal{F}$ using the augmented input $[U_1, V_1, \ldots, U_p, V_p]$ and $\boldsymbol{y}$.

(d) **for** $j = 1, \ldots, p$, **do**:

(1) For each sub-feature k, estimate the local-quadratic smoothed Permutation SHAP derivatives $\widehat{\psi}^{\pm}_{j,k}(\boldsymbol{x}_t)$ for each t. Aggregate them and get $\widehat{\boldsymbol{\psi}}_j^{\pm} = \frac{1}{n}\sum_{t=1}^n \widehat{\boldsymbol{\psi}}_j^{\pm}(\boldsymbol{x}_t)$.

(2) Compute the neural mirror statistic $M_j = \operatorname{sign}\Big(\langle \widehat{\boldsymbol{\psi}}_j^{+}, \widehat{\boldsymbol{\psi}}_j^{-}\rangle\Big)\Big(\|\widehat{\boldsymbol{\psi}}_j^{+}\|_2 + \|\widehat{\boldsymbol{\psi}}_j^{-}\|_2\Big)$.

(e) **end for**

(f) Compute the mirror threshold $\tau_q = \min\Big\{t > 0 : \frac{\#\{j:M_j\le -t\}+1}{\#\{j:M_j\ge t\}\vee 1} \le q\Big\}$.

(g) Return $\widehat{\mathcal{S}}_1 = \{j : M_j \ge \tau_q\}$.

B Flowchart of GGM and PSGM

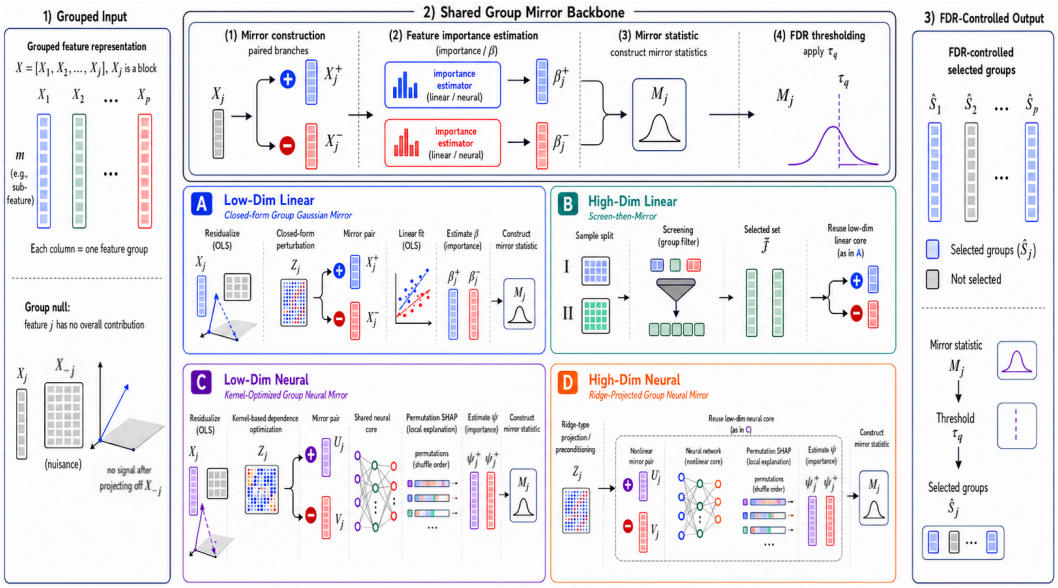


Figure 1: Flowchart of GGM and PSGM

C Proof of Theorems and Propositions

C.1 Proof of Proposition 1

Proposition 1. *In a grouped design $X = [X_1, \ldots, X_p]$ with $X_j \in \mathbb{R}^{n\times m}$ and $\boldsymbol{\beta}_j \in \mathbb{R}^m$, applying the Gaussian Mirror separately to each sub-feature of X_j targets the coordinate-wise nulls $H_{0,(j,\ell)}$: $\beta_{j,\ell} = 0$ for $\ell = 0, \ldots, m-1$, rather than the grouped null $H_{0,j} : \boldsymbol{\beta}_j = (\beta_{j,0}, \ldots, \beta_{j,m-1}) = \mathbf{0}$.*

Proof. In the grouped design, the j-th original feature is represented by the block

$$X_j = [x_{j,0}, x_{j,1}, \ldots, x_{j,m-1}] \in \mathbb{R}^{n\times m},$$

with coefficient vector $\boldsymbol{\beta}_j = (\beta_{j,0}, \beta_{j,1}, \ldots, \beta_{j,m-1})^\top \in \mathbb{R}^m$. Therefore, the feature-level null hypothesis for group j is $H_{0,j} : \boldsymbol{\beta}_j = \mathbf{0}$, which is equivalent to the joint condition

$$\beta_{j,0} = \beta_{j,1} = \cdots = \beta_{j,m-1} = 0.$$

If the original Gaussian Mirror is applied separately to each sub-feature $x_{j,\ell}$ of X_j, then each mirror construction is associated with a single scalar coefficient $\beta_{j,\ell}$. Thus the null hypothesis targeted by the ℓ-th coordinate-wise mirror construction is

$$H_{0,(j,\ell)} : \beta_{j,\ell} = 0, \qquad \ell = 0, \ldots, m-1.$$

This coordinate-wise null hypothesis is not the same inferential object as the grouped null $H_{0,j}$: $\boldsymbol{\beta}_j = \mathbf{0}$, which concerns the simultaneous vanishing of all coefficients in the block.

Consequently, applying the original Gaussian Mirror coordinate by coordinate targets the collection of coordinate-wise hypotheses

$$\{H_{0,(j,\ell)} : \ell = 0, \ldots, m-1\},$$

rather than the grouped feature-level null $H_{0,j}$. This completes the proof. □

C.2 Proof of Theorem 1

Theorem 1. *Assume that $R_j^\top R_j$ and $Q_j^\top Q_j$ are non-singular. For a grouped linear model with $X_j^\pm = X_j \pm \widetilde{Z}_j G_j$, the coefficient vectors $\widehat{\boldsymbol{\beta}}_j^+$ and $\widehat{\boldsymbol{\beta}}_j^-$ satisfy $\mathbb{E}[\widehat{\boldsymbol{\beta}}_j^+] = \mathbb{E}[\widehat{\boldsymbol{\beta}}_j^-] = \mathbf{0}$, $\mathrm{Cov}(\widehat{\boldsymbol{\beta}}_j^+) =$*

$\mathrm{Cov}(\widehat{\boldsymbol{\beta}}_j^-)$, $\mathrm{Cov}(\widehat{\boldsymbol{\beta}}_j^+, \widehat{\boldsymbol{\beta}}_j^-) = 0$*, and are jointly Gaussian under* $H_{0,j}$*, which yields null symmetry of the block-level mirror statistic* M_j*. Alternatively,* $\mathbb{E}[\widehat{\boldsymbol{\beta}}_j^+] = \mathbb{E}[\widehat{\boldsymbol{\beta}}_j^-] = \frac{1}{2}\boldsymbol{\beta}_j$ *under* $H_{1,j}$*.*

Proof. Fix a group j. We have $R_j = M_{X_{-j}} X_j$, $Q_j = (I_n - P_{[X_{-j}, R_j]}) Z_j$, $\widetilde{Z}_j = Q_j (Q_j^\top Q_j)^{-1/2}$ and $G_j = (R_j^\top R_j)^{1/2}$. The nonsingularity of $R_j^\top R_j$ and $Q_j^\top Q_j$ guarantees that G_j and $\widetilde{Z}_j$ are well-defined. Since Q_j is obtained by projecting Z_j onto the orthogonal complement of the column space of $[X_{-j}, R_j]$, we have

$$Q_j^\top X_{-j} = 0, \qquad Q_j^\top R_j = 0.$$

By the definition of $\widetilde{Z}_j$, we have

$$\widetilde{Z}_j^\top \widetilde{Z}_j = (Q_j^\top Q_j)^{-1/2} Q_j^\top Q_j (Q_j^\top Q_j)^{-1/2} = I_m.$$

$$\widetilde{Z}_j^\top X_{-j} = 0, \qquad \widetilde{Z}_j^\top R_j = 0.$$

Define the residualized mirror blocks $V_j^\pm = M_{X_{-j}} X_j^\pm$. Since $X_j^\pm = X_j \pm \widetilde{Z}_j G_j$ and $\widetilde{Z}_j$ is orthogonal to the column space of X_{-j}, we have

$$M_{X_{-j}} \widetilde{Z}_j = \widetilde{Z}_j.$$

Therefore,

$$V_j^\pm = M_{X_{-j}} X_j^\pm = R_j \pm \widetilde{Z}_j G_j,$$

We first establish the key block orthogonality. Since $G_j = (R_j^\top R_j)^{1/2}$, we have $G_j^\top = G_j$ and $G_j^2 = R_j^\top R_j$. Using $R_j^\top \widetilde{Z}_j = 0$ and $\widetilde{Z}_j^\top \widetilde{Z}_j = I_m$, we obtain

$$\begin{aligned}
(V_j^+)^\top V_j^- &= (R_j + \widetilde{Z}_j G_j)^\top (R_j - \widetilde{Z}_j G_j) \\
&= R_j^\top R_j - R_j^\top \widetilde{Z}_j G_j + G_j \widetilde{Z}_j^\top R_j - G_j \widetilde{Z}_j^\top \widetilde{Z}_j G_j \\
&= R_j^\top R_j - G_j^2 \\
&= 0_{m \times m}.
\end{aligned}$$

Similarly,

$$\begin{aligned}
(V_j^+)^\top V_j^+ &= (R_j + \widetilde{Z}_j G_j)^\top (R_j + \widetilde{Z}_j G_j) \\
&= R_j^\top R_j + G_j^2 \\
&= 2 R_j^\top R_j,
\end{aligned}$$

and

$$(V_j^-)^\top V_j^- = 2 R_j^\top R_j.$$

Now consider the grouped null $H_{0,j} : \boldsymbol{\beta}_j = \mathbf{0}$. Under this null, the grouped linear model can be written as

$$Y = X_{-j} \boldsymbol{\beta}_{-j} + \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2 I_n).$$

Residualizing with respect to X_{-j} gives

$$Y^* := M_{X_{-j}} Y = M_{X_{-j}} \varepsilon =: \varepsilon^*.$$

Hence $\mathbb{E}[\varepsilon^*] = 0$ and $\mathrm{Cov}(\varepsilon^*) = \sigma^2 M_{X_{-j}}$. By the Frisch–Waugh–Lovell theorem, the fitted coefficients of X_j^+ and X_j^- in the regression on $[X_{-j}, X_j^+, X_j^-]$ are equal to the fitted coefficients from regressing Y^* on the residualized blocks $[V_j^+, V_j^-]$. Since

$$(V_j^+)^\top V_j^- = 0_{m \times m},$$

the two residualized mirror blocks are block-orthogonal. Therefore,

$$\widehat{\boldsymbol{\beta}}_j^+ = \big((V_j^+)^\top V_j^+\big)^{-1} (V_j^+)^\top Y^* = (2 R_j^\top R_j)^{-1} (V_j^+)^\top \varepsilon^*,$$

and

$$\widehat{\boldsymbol{\beta}}_j^- = \big((V_j^-)^\top V_j^-\big)^{-1} (V_j^-)^\top Y^* = (2 R_j^\top R_j)^{-1} (V_j^-)^\top \varepsilon^*.$$

Since $\mathbb{E}[\varepsilon^*] = 0$, it follows that

$$\mathbb{E}[\widehat{\boldsymbol{\beta}}_j^+] = \mathbb{E}[\widehat{\boldsymbol{\beta}}_j^-] = \mathbf{0}.$$

Moreover, V_j^+ and V_j^- lie in the residual space of X_{-j}. Hence $M_{X_{-j}} V_j^\pm = V_j^\pm$. Thus,

$$\begin{aligned}
\mathrm{Cov}(\widehat{\boldsymbol{\beta}}_j^+) &= (2R_j^\top R_j)^{-1}(V_j^+)^\top \mathrm{Cov}(\varepsilon^*) V_j^+ (2R_j^\top R_j)^{-1} \\
&= \sigma^2 (2R_j^\top R_j)^{-1}(V_j^+)^\top M_{X_{-j}} V_j^+ (2R_j^\top R_j)^{-1} \\
&= \sigma^2 (2R_j^\top R_j)^{-1}(V_j^+)^\top V_j^+ (2R_j^\top R_j)^{-1} \\
&= \sigma^2 (2R_j^\top R_j)^{-1}(2R_j^\top R_j)(2R_j^\top R_j)^{-1} \\
&= \sigma^2 (2R_j^\top R_j)^{-1}.
\end{aligned}$$

The same calculation gives

$$\mathrm{Cov}(\widehat{\boldsymbol{\beta}}_j^-) = \sigma^2 (2R_j^\top R_j)^{-1}.$$

Therefore,

$$\mathrm{Cov}(\widehat{\boldsymbol{\beta}}_j^+) = \mathrm{Cov}(\widehat{\boldsymbol{\beta}}_j^-) = \sigma^2 (2R_j^\top R_j)^{-1}.$$

For the cross-covariance, we have

$$\begin{aligned}
\mathrm{Cov}(\widehat{\boldsymbol{\beta}}_j^+, \widehat{\boldsymbol{\beta}}_j^-) &= (2R_j^\top R_j)^{-1}(V_j^+)^\top \mathrm{Cov}(\varepsilon^*) V_j^- (2R_j^\top R_j)^{-1} \\
&= \sigma^2 (2R_j^\top R_j)^{-1}(V_j^+)^\top M_{X_{-j}} V_j^- (2R_j^\top R_j)^{-1} \\
&= \sigma^2 (2R_j^\top R_j)^{-1}(V_j^+)^\top V_j^- (2R_j^\top R_j)^{-1} \\
&= 0_{m\times m}.
\end{aligned}$$

Furthermore, $\widehat{\boldsymbol{\beta}}_j^+$ and $\widehat{\boldsymbol{\beta}}_j^-$ are linear transformations of the Gaussian noise vector ε. Hence they are jointly Gaussian. Since jointly Gaussian random vectors with zero cross-covariance are independent, we conclude that $\widehat{\boldsymbol{\beta}}_j^+$ and $\widehat{\boldsymbol{\beta}}_j^-$ are independent centered Gaussian vectors with identical covariance.

We now verify the null symmetry of the block-level mirror statistic. Recall that

$$M_j = \mathrm{sign}\Big(\langle \widehat{\boldsymbol{\beta}}_j^+, \widehat{\boldsymbol{\beta}}_j^- \rangle\Big) \left(\|\widehat{\boldsymbol{\beta}}_j^+\|_2 + \|\widehat{\boldsymbol{\beta}}_j^-\|_2 \right).$$

Since $\widehat{\boldsymbol{\beta}}_j^-$ is a centered Gaussian vector, $\widehat{\boldsymbol{\beta}}_j^- \overset{d}{=} -\widehat{\boldsymbol{\beta}}_j^-$. Together with independence, this gives

$$(\widehat{\boldsymbol{\beta}}_j^+, \widehat{\boldsymbol{\beta}}_j^-) \overset{d}{=} (\widehat{\boldsymbol{\beta}}_j^+, -\widehat{\boldsymbol{\beta}}_j^-).$$

We define $\mathcal{M}(a,b) = \mathrm{sign}(\langle a,b\rangle)(\|a\|_2 + \|b\|_2)$, then $\mathcal{M}(a,-b) = \mathrm{sign}(-\langle a,b\rangle)(\|a\|_2 + \|b\|_2) = -\mathcal{M}(a,b)$, with the convention $\mathrm{sign}(0) = 0$. Therefore,

$$M_j = \mathcal{M}(\widehat{\boldsymbol{\beta}}_j^+, \widehat{\boldsymbol{\beta}}_j^-) \overset{d}{=} \mathcal{M}(\widehat{\boldsymbol{\beta}}_j^+, -\widehat{\boldsymbol{\beta}}_j^-) = -M_j.$$

Thus M_j is symmetric about zero under $H_{0,j}$. Now, it only remains to verify the alternative mean statement. Under $H_{1,j} : \boldsymbol{\beta}_j \neq \mathbf{0}$, the model can be written as

$$Y = X_{-j}\boldsymbol{\beta}_{-j} + X_j \boldsymbol{\beta}_j + \varepsilon.$$

$$Y^* = M_{X_{-j}} Y = R_j \boldsymbol{\beta}_j + \varepsilon^*.$$

Using $V_j^\pm = R_j \pm \widetilde{Z}_j G_j$, we obtain

$$R_j = \frac{1}{2}(V_j^+ + V_j^-).$$

$$R_j \boldsymbol{\beta}_j = V_j^+ \frac{\boldsymbol{\beta}_j}{2} + V_j^- \frac{\boldsymbol{\beta}_j}{2}.$$

Thus, in the residualized regression of Y^* on $[V_j^+, V_j^-]$, the population coefficients on the two mirror blocks are both $\boldsymbol{\beta}_j/2$. Since the noise has mean zero, the OLS fitted coefficient vectors satisfy

$$\mathbb{E}[\widehat{\boldsymbol{\beta}}_j^+] = \mathbb{E}[\widehat{\boldsymbol{\beta}}_j^-] = \frac{1}{2}\boldsymbol{\beta}_j.$$

This completes the proof. □

C.3 Proof of Proposition 2

Proposition 2. *Given the properties in Theorem 1, Group Gaussian Mirror asymptotically controls the FDR at the target level q.*

Proof. Let S_0 and S_1 denote the sets of null and non-null grouped features, respectively. For any threshold $t > 0$, define

$$R(t) = \#\{j : M_j \geq t\}, \quad V(t) = \#\{j \in S_0 : M_j \geq t\},$$
$$V^-(t) = \#\{j \in S_0 : M_j \leq -t\}, \quad N^-(t) = \#\{j : M_j \leq -t\}.$$

Here $R(t)$ is the total number of selected grouped features at threshold t, while $V(t)$ is the number of false discoveries. The realized false discovery proportion is

$$\mathrm{FDP}(t) = \frac{V(t)}{R(t) \vee 1}.$$

The Group Gaussian Mirror procedure estimates the FDP by $\widehat{\mathrm{FDP}}(t) = \frac{1+N^-(t)}{R(t)\vee 1}$ and chooses

$$\tau_q = \inf\left\{t > 0 : \widehat{\mathrm{FDP}}(t) \leq q\right\}.$$

We first use the null symmetry property. For every null grouped feature $j \in S_0$,

$$M_j \stackrel{d}{=} -M_j.$$

Consequently, for every $t > 0$,

$$\mathbb{P}(M_j \geq t) = \mathbb{P}(M_j \leq -t), \quad \mathbb{E}[V(t)] = \mathbb{E}[V^-(t)].$$

Thus, at the population level, the positive null tail and the negative null tail contain the same expected number of grouped features. To transfer this expectation-level equality to empirical tail counts, we impose the standard aggregate weak-dependence condition used in Gaussian-Mirror-type and data-splitting FDR proofs. Let $p_0 = |S_0|$. Assume that, for some constants $C > 0$ and $\alpha \in (0, 2)$,

$$\sup_{s\in\mathbb{R}} \mathrm{Var}\left(\sum_{j\in S_0} \mathbf{1}\{M_j > s\}\right) \leq C p_0^{\alpha}.$$

This condition only requires that the dependence among the null mirror-tail indicators is not so dense that their aggregate variance has order p_0^2. In particular, strong within-group dependence, and even strong dependence among some null groups, are allowed provided that the overall dependence is sufficiently sparse or weak in aggregate. We now show that this variance condition implies uniform concentration of the null tails. Define

$$\widehat{G}_+(t) = \frac{V(t)}{p_0}, \qquad G_+(t) = \frac{\mathbb{E}[V(t)]}{p_0}.$$

For every fixed t, Chebyshev's inequality gives

$$\mathbb{P}\left(\left|\widehat{G}_+(t) - G_+(t)\right| > \varepsilon\right) \leq \frac{C p_0^{\alpha}}{\varepsilon^2 p_0^2} = \frac{C}{\varepsilon^2} p_0^{\alpha-2} \longrightarrow 0.$$

Because both $\widehat{G}_+(t)$ and $G_+(t)$ are monotone, the usual finite-grid and monotonicity yields

$$\sup_{t>0}\left|\frac{V(t) - \mathbb{E}[V(t)]}{p_0}\right| = o_p(1). \quad \sup_{t>0}\left|\frac{V^-(t) - \mathbb{E}[V^-(t)]}{p_0}\right| = o_p(1).$$

Combining these two uniform concentration statements with $\mathbb{E}[V(t)] = \mathbb{E}[V^-(t)]$ gives

$$\sup_{t>0} \frac{|V(t) - V^-(t)|}{p_0} = o_p(1).$$

Therefore, the empirical positive and negative null tails are uniformly asymptotically balanced. It remains to evaluate this relation at the data-dependent threshold τ_q. We impose the standard non-degeneracy condition that there exist constants $T > 0$ and $c > 0$ such that

$$\mathbb{P}(\tau_q \leq T) \longrightarrow 1, \quad \mathbb{P}\big(R(T) \geq c p_0\big) \longrightarrow 1.$$

This condition ensures that the procedure admits a feasible threshold and makes a non-vanishing number of discoveries. Since $R(t)$ is non-increasing in t, on the event $\{\tau_q \le T\}$, we have $R(\tau_q) \ge R(T)$. Hence, with probability tending to one, $R(\tau_q) \ge cp_0$. Using the uniform null-tail balance, we obtain

$$\frac{|V(\tau_q) - V^-(\tau_q)|}{R(\tau_q) \vee 1} \le \frac{\sup_{0<t\le T} |V(t) - V^-(t)|}{cp_0} = o_p(1).$$

It follows that

$$\mathrm{FDP}(\tau_q) = \frac{V(\tau_q)}{R(\tau_q) \vee 1} \le \frac{V^-(\tau_q)}{R(\tau_q) \vee 1} + o_p(1).$$

Since every negative null statistic is also counted by the total negative tail, $V^-(t) \le N^-(t)$, for every $t > 0$. Therefore,

$$\mathrm{FDP}(\tau_q) \le \frac{N^-(\tau_q)}{R(\tau_q) \vee 1} + o_p(1) \le \frac{1 + N^-(\tau_q)}{R(\tau_q) \vee 1} + o_p(1).$$

By the definition of τ_q,

$$\frac{1 + N^-(\tau_q)}{R(\tau_q) \vee 1} \le q.$$

$$\mathrm{FDP}(\tau_q) \le q + o_p(1).$$

Finally, since $0 \le \mathrm{FDP}(\tau_q) \le 1$, boundedness implies uniform integrability. Thus,

$$\limsup_{p_0\to\infty} \mathrm{FDR} = \limsup_{p_0\to\infty} \mathbb{E}\big[\mathrm{FDP}(\tau_q)\big] \le q.$$

Therefore, under null symmetry, the aggregate weak-dependence condition, and the non-degenerate-threshold condition, the Group Gaussian Mirror procedure asymptotically controls the false discovery rate at the target level q. For the high-dimensional two-stage procedure, the same argument is applied conditionally on the first-stage screened set. On the sure-screening event, the second-stage statistics for retained null grouped features inherit the low-dimensional null symmetry, while unretained features have mirror statistic zero and do not enter either tail for $t > 0$. If the aggregate variance and non-degeneracy conditions hold conditionally for the second-stage mirror statistics and the sure-screening probability tends to one, then the same proof yields success asymptotic FDR control. This completes the proof. □

C.4 Proof of Proposition 3

Proposition 3. *For a grouped linear model with gaussian noise $\varepsilon \sim N(0, \sigma^2\Sigma)$, where $\Sigma \succ 0$, let L satisfy $L^\top L = \Sigma^{-1}$, define $\bar{Y} = LY$ and $\bar{X}_k = LX_k$ for $k = 1, \dots, p$. Then applying the OLS Group Gaussian Mirror to the whitened system $(\bar{Y}, \bar{X}, \bar{\varepsilon})$ is equivalent to applying GLS Group Gaussian Mirror to the original system (Y, X, ε) with Σ^{-1}-inner product.*

Proof. Consider the grouped linear model

$$Y = \sum_{k=1}^{p} X_k \boldsymbol{\beta}_k + \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2\Sigma),$$

where $\Sigma \succ 0$. Let L satisfy $L^\top L = \Sigma^{-1}$, following the above definition, we have

$$\bar{Y} = LY = \sum_{k=1}^{p} LX_k \boldsymbol{\beta}_k + L\varepsilon = \sum_{k=1}^{p} \bar{X}_k \boldsymbol{\beta}_k + \bar{\varepsilon}.$$

Moreover, $\mathbb{E}[\bar{\varepsilon}] = 0$ and $\mathrm{Cov}(\bar{\varepsilon}) = L\,\mathrm{Cov}(\varepsilon)L^\top = \sigma^2 L\Sigma L^\top$. Since L is non-singular and

$$\Sigma = (L^\top L)^{-1} = L^{-1}L^{-\top}.$$

Therefore,

$$L\Sigma L^\top = L(L^{-1}L^{-\top})L^\top = I_n.$$

Hence

$$\bar{\varepsilon} \sim N(0, \sigma^2 I_n).$$

Thus the whitened system is a grouped linear model with spherical Gaussian noise. We then show that OLS in the whitened system is equivalent to GLS in the original system. For any coefficient vector $\boldsymbol{\beta}$, where $\bar{X} = [\bar{X}_1, \ldots, \bar{X}_p]$ and $X = [X_1, \ldots, X_p]$,

$$\|\bar{Y} - \bar{X}\boldsymbol{\beta}\|_2^2 = \|L(Y - X\boldsymbol{\beta})\|_2^2.$$

Expanding the right-hand side gives

$$\|L(Y - X\boldsymbol{\beta})\|_2^2 = (Y - X\boldsymbol{\beta})^\top L^\top L(Y - X\boldsymbol{\beta}) = (Y - X\boldsymbol{\beta})^\top \Sigma^{-1}(Y - X\boldsymbol{\beta}).$$

This is exactly the GLS objective in the original system. Therefore, minimizing the OLS objective after whitening is equivalent to minimizing the GLS objective before whitening.

Now, it only remains to relate the mirror construction in the whitened system to the Σ^{-1}-inner product in the original system. For any two original-scale vectors $a, b \in \mathbb{R}^n$,

$$\langle La, Lb\rangle = (La)^\top (Lb) = a^\top L^\top Lb = a^\top \Sigma^{-1} b.$$

Thus the Euclidean inner product after whitening is exactly the Σ^{-1}-inner product before whitening. In particular, the Euclidean projection onto the column space of $\bar{X}_{-j} = LX_{-j}$ is

$$P_{\bar{X}_{-j}} = \bar{X}_{-j}(\bar{X}_{-j}^\top \bar{X}_{-j})^{-1}\bar{X}_{-j}^\top.$$

Substituting $\bar{X}_{-j} = LX_{-j}$, we get

$$P_{\bar{X}_{-j}} = LX_{-j}(X_{-j}^\top L^\top LX_{-j})^{-1}X_{-j}^\top L^\top = LX_{-j}(X_{-j}^\top \Sigma^{-1} X_{-j})^{-1}X_{-j}^\top L^\top.$$

Define the GLS projection in the original system by

$$P_{X_{-j}}^{\Sigma} = X_{-j}(X_{-j}^\top \Sigma^{-1} X_{-j})^{-1}X_{-j}^\top \Sigma^{-1}.$$

Since $\Sigma^{-1} = L^\top L$, we have

$$LP_{X_{-j}}^{\Sigma} = LX_{-j}(X_{-j}^\top \Sigma^{-1} X_{-j})^{-1}X_{-j}^\top \Sigma^{-1} = LX_{-j}(X_{-j}^\top \Sigma^{-1} X_{-j})^{-1}X_{-j}^\top L^\top L.$$

On the other hand, we also have

$$P_{\bar{X}_{-j}} L = LX_{-j}(X_{-j}^\top \Sigma^{-1} X_{-j})^{-1}X_{-j}^\top L^\top L.$$

Therefore, $P_{\bar{X}_{-j}} L = LP_{X_{-j}}^{\Sigma}$. Consequently, if $M_{\bar{X}_{-j}} = I_n - P_{\bar{X}_{-j}}$ and $M_{X_{-j}}^{\Sigma} = I_n - P_{X_{-j}}^{\Sigma}$, then $M_{\bar{X}_{-j}} L = LM_{X_{-j}}^{\Sigma}$. The same correspondence applies to the residualized group block. In the whitened system,

$$\bar{R}_j = M_{\bar{X}_{-j}} \bar{X}_j = M_{\bar{X}_{-j}} LX_j = LM_{X_{-j}}^{\Sigma} X_j.$$

Thus, if $R_j^{\Sigma} = M_{X_{-j}}^{\Sigma} X_j$, we have $\bar{R}_j = LR_j^{\Sigma}$. The whitened Gram matrix is therefore

$$\bar{R}_j^\top \bar{R}_j = (R_j^{\Sigma})^\top L^\top LR_j^{\Sigma} = (R_j^{\Sigma})^\top \Sigma^{-1} R_j^{\Sigma}.$$

Hence the Euclidean Gram matrix in the whitened system is exactly the Σ^{-1}-metric Gram matrix in the original system.

Now consider the mirror noise construction. In the whitened system, the OLS Group Gaussian Mirror projects the noise block away from the column spaces of $\bar{X}_{-j}$ and $\bar{R}_j$, so that the projected noise block $\bar{Q}_j$ satisfies

$$\bar{Q}_j^\top \bar{X}_{-j} = 0, \qquad \bar{Q}_j^\top \bar{R}_j = 0.$$

Writing $\bar{Q}_j = LQ_j$, $\bar{X}_{-j} = LX_{-j}$, and $\bar{R}_j = LR_j^{\Sigma}$, these orthogonality relations become

$$Q_j^\top L^\top LX_{-j} = 0, \qquad Q_j^\top L^\top LR_j^{\Sigma} = 0.$$

Since $L^\top L = \Sigma^{-1}$, this is equivalent to

$$Q_j^\top \Sigma^{-1} X_{-j} = 0, \qquad Q_j^\top \Sigma^{-1} R_j^{\Sigma} = 0.$$

Thus orthogonality after whitening is the same as Σ^{-1}-orthogonality in the original system.

Similarly, the normalization in the whitened system, $\widetilde{\widetilde{Z}}_j^\top \widetilde{\widetilde{Z}}_j = I_m$, corresponds to $\widetilde{Z}_j^\top \Sigma^{-1} \widetilde{Z}_j = I_m$ in the original system, because $\widetilde{\widetilde{Z}}_j = L\widetilde{Z}_j$ implies

$$\widetilde{\widetilde{Z}}_j^\top \widetilde{\widetilde{Z}}_j = \widetilde{Z}_j^\top L^\top L \widetilde{Z}_j = \widetilde{Z}_j^\top \Sigma^{-1} \widetilde{Z}_j.$$

Therefore, the OLS Group Gaussian Mirror construction in the whitened system is identical to the GLS Group Gaussian Mirror construction in the original system with all projections, normalizations, and orthogonality relations computed under the Σ^{-1}-inner product.

Finally, since the whitened error $\bar{\varepsilon} \sim N(0, \sigma^2 I_n)$, the OLS Group Gaussian Mirror applies directly to the whitened system. Since OLS estimation in the whitened system is equivalent to GLS estimation in the original system, applying OLS Group Gaussian Mirror to $(\bar{Y}, \bar{X}, \bar{\varepsilon})$ is equivalent to applying GLS Group Gaussian Mirror to (Y, X, ε) with the Σ^{-1}-inner product. This completes the proof. □

C.5 Proof of Proposition 4

Proposition 4. *For a grouped linear model, assume the weight obtained from the first-stage Group Lasso screener is $\widehat{\boldsymbol{\beta}}^{GL}$, and let $\widehat{\mathcal{A}} = \{j : \widehat{\boldsymbol{\beta}}_j^{GL} \neq \mathbf{0}\}$ be the selected grouped feature set. Then, in general, the event $\widehat{\mathcal{A}} = A$ is not polyhedral in Y.*

Proof. The first-stage Group Lasso estimator is defined by

$$\widehat{\boldsymbol{\beta}}^{GL} \in \arg\min_{\{\boldsymbol{\beta}_j\}_{j=1}^p} \left\{ \frac{1}{2n} \left\| Y - \sum_{j=1}^p X_j \boldsymbol{\beta}_j \right\|_2^2 + \lambda \sum_{j=1}^p w_j \|\boldsymbol{\beta}_j\|_2 \right\}.$$

Let $\widehat{\mathcal{A}} = \{j : \widehat{\boldsymbol{\beta}}_j^{GL} \neq \mathbf{0}\}$ be the selected grouped feature set, and we define the fitted residual $\widehat{r} = Y - \sum_{j=1}^p X_j \widehat{\boldsymbol{\beta}}_j^{GL}$. Then the Karush-Kuhn-Tucker (KKT) conditions for Group Lasso imply that, for each active grouped feature $j \in \widehat{\mathcal{A}}$,

$$X_j^\top \widehat{r} = n\lambda w_j \frac{\widehat{\boldsymbol{\beta}}_j^{GL}}{\|\widehat{\boldsymbol{\beta}}_j^{GL}\|_2},$$

whereas for each inactive grouped feature $j \notin \widehat{\mathcal{A}}$,

$$\|X_j^\top \widehat{r}\|_2 \leq n\lambda w_j.$$

Therefore, the event $\widehat{\mathcal{A}} = A$ is characterized by constraints involving group ℓ_2-norms and, for active grouped features, normalized direction vectors of the form $\widehat{\boldsymbol{\beta}}_j^{GL} / \|\widehat{\boldsymbol{\beta}}_j^{GL}\|_2$. This differs from the coordinate-wise lasso. In the ordinary lasso, the inactive KKT condition is

$$|x_j^\top \widehat{r}| \leq n\lambda,$$

which is equivalent to two linear inequalities. Together with a fixed active sign pattern, the lasso selection event can be represented as a finite intersection of affine half-spaces in Y, and hence is polyhedral. For Group Lasso, however, the inactive grouped feature condition

$$\|X_j^\top \widehat{r}\|_2 \leq n\lambda w_j$$

is a second-order cone constraint whenever the group dimension is larger than one. Such a constraint is generally not representable as a finite intersection of linear inequalities. Moreover, the active grouped feature condition depends on the continuous direction

$$\widehat{\boldsymbol{\beta}}_j^{GL} / \|\widehat{\boldsymbol{\beta}}_j^{GL}\|_2,$$

rather than on a finite sign pattern as in the coordinate-wise lasso. Therefore, the event $\widehat{\mathcal{A}} = A$ induced by Group Lasso is not polyhedral in the response vector. This completes the proof.

Also, this non-polyhedral structure can be seen explicitly in a simple example. Consider a model with a single group $X_1 \in \mathbb{R}^{n\times m}$, where $m > 1$, and suppose $X_1^\top X_1 = nI_m$. In this case, the Group Lasso solution is obtained by group soft-thresholding, and the group is selected if and only if

$$\|X_1^\top Y\|_2 > n\lambda w_1.$$

Equivalently, the non-selection event is

$$\|X_1^\top Y\|_2 \le n\lambda w_1.$$

This is the inverse image of an m-dimensional Euclidean ball under the linear map $Y \mapsto X_1^\top Y$. For $m > 1$, this set has a curved boundary and is not a polyhedron in Y. □

C.6 Proof of Theorem 2

Theorem 2. *Assume that: (i) the two data chunks share the same grouped-feature null structure and are independent, or asymptotically independent; (ii) the first-stage screening procedure satisfies a sure screening property, namely $\mathbb{P}(\mathcal{S}_1 \subseteq \widehat{\mathcal{A}}) \to 1$; and (iii) with probability tending to one, the second-stage restricted design is low-dimensional and full-rank. Then for every null grouped feature $j \in \mathcal{S}_0$, the two-stage mirror statistic $\widetilde{M}_j$ satisfies*

$$\sup_{t>0} \left| \mathbb{P}(\widetilde{M}_j > t) - \mathbb{P}(\widetilde{M}_j < -t) \right| \to 0.$$

Proof. Fix a null grouped feature $j \in \mathcal{S}_0$. Let $\widehat{\mathcal{A}}$ denote the first-stage selected grouped feature set. Since $\widehat{\mathcal{A}}$ is computed only from the first data chunk $(X^{(1)}, \boldsymbol{y}^{(1)})$, it is measurable with respect to the first-stage data. Recall that the extended two-stage mirror statistic is defined by

$$\widetilde{M}_j = \begin{cases} M_j, & j \in \widehat{\mathcal{A}}, \\ 0, & j \notin \widehat{\mathcal{A}}, \end{cases}$$

where M_j is the Group Gaussian Mirror statistic computed on the second data chunk within the restricted model indexed by $\widehat{\mathcal{A}}$. For any $t > 0$, by the law of total expectation,

$$\mathbb{P}(\widetilde{M}_j > t) - \mathbb{P}(\widetilde{M}_j < -t) = \mathbb{E}\left[\mathbb{P}(\widetilde{M}_j > t \mid \widehat{\mathcal{A}}) - \mathbb{P}(\widetilde{M}_j < -t \mid \widehat{\mathcal{A}}) \right].$$

We now analyze the conditional difference inside the expectation. Firstly, we consider the event $j \notin \widehat{\mathcal{A}}$. By construction, $\widetilde{M}_j = 0$. Hence, for every $t > 0$,

$$\mathbb{P}(\widetilde{M}_j > t \mid \widehat{\mathcal{A}}) = \mathbb{P}(\widetilde{M}_j < -t \mid \widehat{\mathcal{A}}) = 0.$$

Thus the conditional contribution is exactly zero on $\{j \notin \widehat{\mathcal{A}}\}$. Next, we consider the event $j \in \widehat{\mathcal{A}}$. On this event, $\widetilde{M}_j = M_j$, and M_j is obtained by applying the low-dimensional Group Gaussian Mirror to the second data chunk restricted to the selected grouped features $\widehat{\mathcal{A}}$. We define

$$\mathcal{E}_n = \left\{ \mathcal{S}_1 \subseteq \widehat{\mathcal{A}} \right\} \cap \{\text{the second-stage restricted design is low-dimensional and full-rank}\}.$$

By assumptions (ii) and (iii), we have

$$\mathbb{P}(\mathcal{E}_n) \to 1.$$

On $\mathcal{E}_n$, the screened model contains all nonzero grouped features. Therefore, for $j \in \mathcal{S}_0 \cap \widehat{\mathcal{A}}$, the coefficient block of j remains null in the second-stage restricted model, and no omitted-signal bias is introduced by the screening step. Moreover, by assumption (i), the second data chunk is independent of $\widehat{\mathcal{A}}$, or at least asymptotically independent in the ordered-data case. Hence, conditional on $\widehat{\mathcal{A}}$, the second-stage procedure is equivalent, up to an $o(1)$ perturbation in the asymptotically independent case, to applying the low-dimensional Group Gaussian Mirror to a fixed deterministic restricted model. On $\mathcal{E}_n$, this restricted model satisfies the low-dimensional full-rank conditions required by Theorem 1. Therefore, we have

$$\sup_{t>0} \left| \mathbb{P}(M_j > t \mid \widehat{\mathcal{A}}) - \mathbb{P}(M_j < -t \mid \widehat{\mathcal{A}}) \right| \to 0 \quad \text{on} \quad \mathcal{E}_n.$$

Since $\widetilde{M}_j = M_j$ whenever $j \in \widehat{\mathcal{A}}$, it follows that

$$\sup_{t>0} \left| \mathbb{P}(\widetilde{M}_j > t \mid \widehat{\mathcal{A}}) - \mathbb{P}(\widetilde{M}_j < -t \mid \widehat{\mathcal{A}}) \right| \to 0 \quad \text{on} \quad \{j \in \widehat{\mathcal{A}}\} \cap \mathcal{E}_n.$$

Combining the two cases, define

$$Z_n = \sup_{t>0} \left| \mathbb{P}(\widetilde{M}_j > t \mid \widehat{\mathcal{A}}) - \mathbb{P}(\widetilde{M}_j < -t \mid \widehat{\mathcal{A}}) \right|.$$

Then $0 \le Z_n \le 1$. On the event $j \notin \widehat{\mathcal{A}}$, we have $Z_n = 0$. On the event $\{j \in \widehat{\mathcal{A}}\} \cap \mathcal{E}_n$, the preceding conditional low-dimensional argument gives $Z_n \to 0$. Since $\mathbb{P}(\mathcal{E}_n^c) \to 0$, we conclude that

$$Z_n \xrightarrow{p} 0.$$

Because Z_n is uniformly bounded, convergence in probability to zero implies $\mathbb{E}[Z_n] \to 0$. Finally,

$$\begin{aligned}
&\sup_{t>0} \left| \mathbb{P}(\widetilde{M}_j > t) - \mathbb{P}(\widetilde{M}_j < -t) \right| \\
&\quad = \sup_{t>0} \left| \mathbb{E}\left[\mathbb{P}(\widetilde{M}_j > t \mid \widehat{\mathcal{A}}) - \mathbb{P}(\widetilde{M}_j < -t \mid \widehat{\mathcal{A}}) \right] \right| \\
&\quad \le \mathbb{E}\left[\sup_{t>0} \left| \mathbb{P}(\widetilde{M}_j > t \mid \widehat{\mathcal{A}}) - \mathbb{P}(\widetilde{M}_j < -t \mid \widehat{\mathcal{A}}) \right| \right] \\
&\quad = \mathbb{E}[Z_n] \to 0.
\end{aligned}$$

This completes the proof. □

C.7 Proof of Proposition 5

Proposition 5. *For a grouped linear model, let $\Phi_j(\boldsymbol{x})$ be the group-level Permutation SHAP value of group j. Then $\Phi_j(\boldsymbol{x}) = \boldsymbol{\beta}_j^\top \left(\boldsymbol{x}_j - \mathbb{E}[X'_j] \right)$ and hence $\boldsymbol{\psi}_j(\boldsymbol{x}) = \nabla_{\boldsymbol{x}_j} \Phi_j(\boldsymbol{x}) = \boldsymbol{\beta}_j$, which coincides with the linear-case coefficient vector $\boldsymbol{\beta}_j$.*

Proof. Consider the grouped linear model

$$f(\boldsymbol{x}) = \alpha + \sum_{\ell=1}^{p} \boldsymbol{x}_\ell^\top \boldsymbol{\beta}_\ell, \quad \boldsymbol{x}_\ell \in \mathbb{R}^m.$$

Fix a grouped feature j, a permutation π, and a background draw X'. The two hybrid inputs in the marginal contribution $\Delta_j(\boldsymbol{x}; \pi, X')$ differ only in the coordinates of group j. All groups appearing before j in the permutation are taken from $\boldsymbol{x}$ in both hybrid inputs, while all groups appearing after j are taken from X' in both hybrid inputs. Therefore, by linearity, we have

$$\begin{aligned}
\Delta_j(\boldsymbol{x}; \pi, X') &= f\big(\boldsymbol{x}_{P_j(\pi) \cup \{j\}}, X'_{-(P_j(\pi) \cup \{j\})}\big) - f\big(\boldsymbol{x}_{P_j(\pi)}, X'_{-P_j(\pi)}\big) \\
&= \boldsymbol{\beta}_j^\top \boldsymbol{x}_j - \boldsymbol{\beta}_j^\top X'_j = \boldsymbol{\beta}_j^\top (\boldsymbol{x}_j - X'_j).
\end{aligned}$$

This expression does not depend on the permutation π. Taking expectation over Π and X', we obtain

$$\Phi_j(\boldsymbol{x}) = \mathbb{E}_{\Pi, X'}[\Delta_j(\boldsymbol{x}; \Pi, X')] = \boldsymbol{\beta}_j^\top \left(\boldsymbol{x}_j - \mathbb{E}[X'_j] \right).$$

$$\nabla_{\boldsymbol{x}_j} \Phi_j(\boldsymbol{x}) = \nabla_{\boldsymbol{x}_j} \left\{ \boldsymbol{\beta}_j^\top \left(\boldsymbol{x}_j - \mathbb{E}[X'_j] \right) \right\} = \boldsymbol{\beta}_j.$$

By the definition of the block-level Permutation SHAP derivative vector, we have

$$\boldsymbol{\psi}_j(\boldsymbol{x}) = \left(\frac{\partial \Phi_j(\boldsymbol{x})}{\partial x_{j,1}}, \ldots, \frac{\partial \Phi_j(\boldsymbol{x})}{\partial x_{j,m}} \right)^\top = \nabla_{\boldsymbol{x}_j} \Phi_j(\boldsymbol{x}) = \boldsymbol{\beta}_j.$$

This completes the proof. □

C.8 Proof of Theorem 3

Theorem 3. *For a fixed nonlinear model f and a null grouped feature j, let $\widehat{\psi}_{j,k}(x_0)$ denote the local-quadratic smoothed Monte Carlo Permutation SHAP derivative estimator. Under some regularity conditions, we have, $\widehat{\psi}_{j,k}(x_0)$ is symmetric around zero for each sub-feature $x_{j,k}$, namely,*

$$\sup_{t>0}\left|\mathbb{P}\Big(\widehat{\psi}_{j,k}(x_0) > t\Big) - \mathbb{P}\Big(\widehat{\psi}_{j,k}(x_0) < -t\Big)\right| \to 0, \quad \forall k.$$

Proof. Fix a null grouped feature j. We interpret the null at the model level: there exists a measurable function g such that

$$f(\boldsymbol{x}) = g(\boldsymbol{x}_{-j}), \qquad \forall \boldsymbol{x}.$$

Hence the fitted model does not depend on any sub-feature in the grouped block j. We first show that the exact group-level Permutation SHAP value is identically zero. Fix any input $\boldsymbol{x}$, any permutation π, and any background draw X'. Since f does not depend on the grouped feature j, the two hybrid inputs in the marginal contribution have the same value under f:

$$f\big(\boldsymbol{x}_{P_j(\pi)\cup\{j\}}, X'_{-(P_j(\pi)\cup\{j\})}\big) = f\big(\boldsymbol{x}_{P_j(\pi)}, X'_{-P_j(\pi)}\big).$$

Therefore, we have $\Delta_j(\boldsymbol{x};\pi,X') = 0$ for every realization of (π, X'). Taking expectation over (Π, X') gives

$$\Phi_j(\boldsymbol{x}) = \mathbb{E}_{\Pi,X'}[\Delta_j(\boldsymbol{x};\Pi,X')] = 0, \qquad \forall \boldsymbol{x}.$$

Thus Φ_j is the zero function, and every population partial derivative with respect to a sub-feature in group j is also zero:

$$\frac{\partial \Phi_j(\boldsymbol{x})}{\partial x_{j,k}} = 0, \qquad k = 1,\dots,m.$$

It now remains to show that the smoothed Monte Carlo estimator of this derivative is asymptotically symmetric. Fix a sub-feature $x_{j,k}$ and a point x_0. Let

$$\widehat{\Phi}_{j,t} = \frac{1}{M_n}\sum_{q=1}^{M_n} Z_{tq}, \qquad t = 1,\dots,n,$$

denote the Monte Carlo Permutation SHAP estimator at the t-th explained point. Under the model-level null, the exact value satisfies $\Phi_j(\boldsymbol{x}_t) = 0$, so the Monte Carlo variables are centered:

$$\mathbb{E}[Z_{tq} \mid \boldsymbol{x}_t] = 0.$$

Assume, as part of the regularity conditions, that conditional on the explained points $\boldsymbol{x}_1,\dots,\boldsymbol{x}_n$, the variables $\{Z_{tq}\}_{1\le t\le n, 1\le q\le M_n}$ are independent,

$$\mathrm{Var}(Z_{tq} \mid \boldsymbol{x}_t) = \tau^2(x_{t,j,k}),$$

and for some $\delta > 0$,

$$\sup_{t,q}\mathbb{E}\big[|Z_{tq}|^{2+\delta} \mid \boldsymbol{x}_t\big] < \infty.$$

We define $U_t = \frac{x_{t,j,k}-x_0}{h_n}$, $r_t = r(U_t) = (1, U_t, U_t^2)^\top$, and $K_t = K(U_t)$. Then the local-quadratic coefficients $\widehat{b} = (\widehat{b}_0, \widehat{b}_1, \widehat{b}_2)^\top$ solve

$$\widehat{b} = \arg\min_{b\in\mathbb{R}^3}\sum_{t=1}^{n} K_t\left(\widehat{\Phi}_{j,t} - b^\top r_t\right)^2.$$

Let $S_n = \sum_{t=1}^n K_t r_t r_t^\top$ and $T_n = \sum_{t=1}^n K_t r_t \widehat{\Phi}_{j,t}$, the normal equations give $\widehat{b} = S_n^{-1}T_n$. Therefore, the local-quadratic derivative estimator can be written as

$$\widehat{\psi}_{j,k}(x_0) = \frac{\widehat{b}_1}{h_n} = \frac{1}{h_n}e_2^\top S_n^{-1}T_n, \qquad e_2 = (0,1,0)^\top.$$

Or equivalently,

$$\widehat{\psi}_{j,k}(x_0) = \sum_{t=1}^{n} a_{nt}\widehat{\Phi}_{j,t}, \qquad a_{nt} = \frac{1}{h_n}e_2^\top S_n^{-1}K_t r_t.$$

Since

$$\widehat{\Phi}_{j,t} = \frac{1}{M_n} \sum_{q=1}^{M_n} Z_{tq},$$

we have the triangular-array representation

$$\widehat{\psi}_{j,k}(x_0) = \sum_{t=1}^{n} \sum_{q=1}^{M_n} \frac{a_{nt}}{M_n} Z_{tq}.$$

Firstly, we compute the conditional mean. Because $\mathbb{E}[Z_{tq} \mid \boldsymbol{x}_t] = 0$, we have

$$\mathbb{E}[\widehat{\Phi}_{j,t} \mid \boldsymbol{x}_t] = \frac{1}{M_n} \sum_{q=1}^{M_n} \mathbb{E}[Z_{tq} \mid \boldsymbol{x}_t] = 0.$$

Hence

$$\mathbb{E}[\widehat{\psi}_{j,k}(x_0) \mid \boldsymbol{x}_1, \ldots, \boldsymbol{x}_n] = \sum_{t=1}^{n} a_{nt} \mathbb{E}[\widehat{\Phi}_{j,t} \mid \boldsymbol{x}_t] = 0.$$

Next, we compute the conditional variance. Since the Monte Carlo draws are independent across both t and q, we have $\mathrm{Var}(\widehat{\Phi}_{j,t} \mid \boldsymbol{x}_t) = \tau^2(x_{t,j,k})/M_n$. Therefore,

$$\mathrm{Var}(T_n \mid \boldsymbol{x}_1, \ldots, \boldsymbol{x}_n) = \frac{1}{M_n} \sum_{t=1}^{n} K_t^2 r_t r_t^\top \tau^2(x_{t,j,k}).$$

Using $\widehat{\psi}_{j,k}(x_0) = h_n^{-1} e_2^\top S_n^{-1} T_n$, we can obtain that

$$\mathrm{Var}(\widehat{\psi}_{j,k}(x_0) \mid \boldsymbol{x}_1, \ldots, \boldsymbol{x}_n) = \frac{1}{h_n^2} e_2^\top S_n^{-1} \left[\frac{1}{M_n} \sum_{t=1}^{n} K_t^2 r_t r_t^\top \tau^2(x_{t,j,k}) \right] S_n^{-1} e_2.$$

Define $\Gamma_n = \frac{1}{nh_n} \sum_{t=1}^{n} K_t r_t r_t^\top$ and $\Omega_n = \frac{1}{nh_n} \sum_{t=1}^{n} K_t^2 r_t r_t^\top \tau^2(x_{t,j,k})$, then we have

$$S_n = nh_n \Gamma_n, \qquad \sum_{t=1}^{n} K_t^2 r_t r_t^\top \tau^2(x_{t,j,k}) = nh_n \Omega_n.$$

Substituting these identities gives

$$\mathrm{Var}(\widehat{\psi}_{j,k}(x_0) \mid \boldsymbol{x}_1, \ldots, \boldsymbol{x}_n) = \frac{1}{M_n nh_n^3} e_2^\top \Gamma_n^{-1} \Omega_n \Gamma_n^{-1} e_2.$$

Under the local design regularity conditions, $\Gamma_n \to \Gamma$ and $\Omega_n \to \Omega$ with Γ positive definite. Hence

$$\mathrm{Var}(\widehat{\psi}_{j,k}(x_0) \mid \boldsymbol{x}_1, \ldots, \boldsymbol{x}_n) = \frac{1}{M_n nh_n^3} V_{j,k}(x_0) + o\left(\frac{1}{M_n nh_n^3} \right),$$

where

$$V_{j,k}(x_0) = e_2^\top \Gamma^{-1} \Omega \Gamma^{-1} e_2.$$

We now prove asymptotic normality. We define $\xi_{ntq} = \frac{a_{nt}}{M_n} Z_{tq}$ and $\widehat{\psi}_{j,k}(x_0) = \sum_{t=1}^{n} \sum_{q=1}^{M_n} \xi_{ntq}$. Conditional on the explained points, the variables ξ_{ntq} are independent and centered. Since

$$S_n^{-1} = (nh_n)^{-1} \Gamma_n^{-1},$$

we have

$$a_{nt} = \frac{1}{h_n} e_2^\top (nh_n)^{-1} \Gamma_n^{-1} K_t r_t = \frac{1}{nh_n^2} e_2^\top \Gamma_n^{-1} K_t r_t.$$

Because K is bounded and compactly supported, r_t is bounded on the support of K. Since Γ_n^{-1} is bounded for all large n, it should follow that

$$\max_{1 \le t \le n} |a_{nt}| = O\left(\frac{1}{nh_n^2} \right).$$

We further let $s_n^2 = \mathrm{Var}(\widehat{\psi}_{j,k}(x_0) \mid \boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)$. From the variance calculation above, we know that

$$s_n^2 \asymp \frac{1}{M_n n h_n^3}.$$

To verify Lyapunov's condition, we use the moment bound. There exists $C > 0$ such that

$$\sup_{t,q} \mathbb{E}[|Z_{tq}|^{2+\delta} \mid \boldsymbol{x}_t] \le C.$$

Therefore,

$$\sum_{t=1}^{n}\sum_{q=1}^{M_n} \mathbb{E}[|\xi_{ntq}|^{2+\delta} \mid \boldsymbol{x}_1, \ldots, \boldsymbol{x}_n] \le C \sum_{t=1}^{n} M_n \left(\frac{|a_{nt}|}{M_n}\right)^{2+\delta}.$$

Only observations with U_t in the support of K contribute. By the local design regularity, the number of such observations is $O(nh_n)$. Hence

$$\sum_{t=1}^{n}\sum_{q=1}^{M_n} \mathbb{E}[|\xi_{ntq}|^{2+\delta} \mid \boldsymbol{x}_1, \ldots, \boldsymbol{x}_n] = O\left(nh_n \cdot M_n \left(\frac{1}{M_n n h_n^2}\right)^{2+\delta}\right) = O\left(\frac{1}{M_n^{1+\delta} n^{1+\delta} h_n^{3+2\delta}}\right).$$

On the other hand, we have

$$s_n^{2+\delta} \asymp \left(\frac{1}{M_n n h_n^3}\right)^{1+\delta/2} = \frac{1}{M_n^{1+\delta/2} n^{1+\delta/2} h_n^{3+3\delta/2}}.$$

Since $h_n \to 0$, $nh_n^3 \to \infty$, and hence $nh_n \to \infty$, we have

$$\frac{\sum_{t=1}^{n}\sum_{q=1}^{M_n} \mathbb{E}[|\xi_{ntq}|^{2+\delta} \mid \boldsymbol{x}_1, \ldots, \boldsymbol{x}_n]}{s_n^{2+\delta}} = O\left(\frac{1}{(M_n n h_n)^{\delta/2}}\right) \to 0,$$

Thus, the Lyapunov's condition holds. Then, by Lyapunov's central limit theorem for triangular arrays, conditional on the explained points, we obtain

$$\frac{\widehat{\psi}_{j,k}(x_0)}{s_n} \xrightarrow{d} N(0,1), \quad \text{or equivalently,} \quad \sqrt{M_n n h_n^3}\, \widehat{\psi}_{j,k}(x_0) \xrightarrow{d} N(0, V_{j,k}(x_0)).$$

Till now, it only remains to translate asymptotic normality into null symmetry. Since $s_n > 0$, for any $t > 0$, we have

$$\mathbb{P}(\widehat{\psi}_{j,k}(x_0) > t) = \mathbb{P}\left(\frac{\widehat{\psi}_{j,k}(x_0)}{s_n} > \frac{t}{s_n}\right) \quad \text{and} \quad \mathbb{P}(\widehat{\psi}_{j,k}(x_0) < -t) = \mathbb{P}\left(\frac{\widehat{\psi}_{j,k}(x_0)}{s_n} < -\frac{t}{s_n}\right).$$

The standardized statistic converges in distribution to $N(0,1)$, whose law is symmetric about zero. Since the limiting distribution function is continuous, the convergence of distribution functions should be uniform. Therefore, we have

$$\sup_{u>0} \left| \mathbb{P}\left(\frac{\widehat{\psi}_{j,k}(x_0)}{s_n} > u\right) - \mathbb{P}\left(\frac{\widehat{\psi}_{j,k}(x_0)}{s_n} < -u\right) \right| \to 0.$$

Taking $u = t/s_n$ yields

$$\sup_{t>0} \left| \mathbb{P}\Big(\widehat{\psi}_{j,k}(x_0) > t\Big) - \mathbb{P}\Big(\widehat{\psi}_{j,k}(x_0) < -t\Big) \right| \to 0.$$

Since $k \in \{1, \ldots, m\}$ was arbitrary, the same conclusion holds coordinate-wise for every sub-feature in the grouped feature block. This completes the proof. □

C.9 Proof of Theorem 4

Theorem 4. *Applying RKHS log-density decomposition and likelihood-ratio score construction to $(U_j(G_j), V_j(G_j), W_j)$ gives the grouped Neural Gaussian Mirror objective:*

$$I_j^K(G_j)^2 = \frac{1}{n^2} \left[(H_n K_U(G_j) H_n) \circ (H_n K_V(G_j) H_n) \circ K_W\right]_{++},$$

where K_U, K_V and K_W are kernel matrices computed from the rows of $U_j(G_j)$, $V_j(G_j)$ and W_j.

Proof. Fix a grouped feature j. For a given perturbation matrix $G_j \in \mathbb{R}^{m\times m}$, we define as before

$$U_j(G_j) = X_j + \widetilde{Z}_j G_j, \qquad V_j(G_j) = X_j - \widetilde{Z}_j G_j, \qquad W_j = X_{-j}.$$

For the i-th observation, we write

$$t_i(G_j) = \big(u_i(G_j), v_i(G_j), w_i\big),$$

where $u_i(G_j)$, $v_i(G_j)$, and w_i denote the i-th rows of $U_j(G_j)$, $V_j(G_j)$, and W_j, respectively. The Neural Gaussian Mirror construction tests the conditional-independence target

$$U_j(G_j) \perp V_j(G_j) \mid W_j.$$

Let $\eta(u, v, w) = \log p_{U,V,W}(u, v, w)$ be the log-density of the triplet $(U_j(G_j), V_j(G_j), W_j)$. As in the Neural Gaussian Mirror construction, we place η in the tensor-product RKHS:

$$\mathcal{H} = \mathcal{H}_U \otimes \mathcal{H}_V \otimes \mathcal{H}_W.$$

Decompose each marginal RKHS into its constant and centered components, $\mathcal{H}_A = \mathcal{H}_A^0 \oplus \mathcal{H}_A^1$ for $A \in \{U, V, W\}$, we then have

$$\mathcal{H} = \bigoplus_{a,b,c\in\{0,1\}} \mathcal{H}_U^a \otimes \mathcal{H}_V^b \otimes \mathcal{H}_W^c.$$

The conditional-independence null space should consist of components involving only main effects and interactions with W_j, but no joint interaction between U_j and V_j. Hence the alternative space is

$$\mathcal{H}_1 = (\mathcal{H}_U^1 \otimes \mathcal{H}_V^1 \otimes \mathcal{H}_W^0) \oplus (\mathcal{H}_U^1 \otimes \mathcal{H}_V^1 \otimes \mathcal{H}_W^1).$$

Let K^1 denote the reproducing kernel of this alternative space. Then the likelihood-ratio score in the alternative direction is the empirical RKHS score

$$S_n(G_j) = \frac{1}{n}\sum_{i=1}^{n} K^1_{t_i(G_j)}.$$

The grouped Neural Gaussian Mirror objective is defined as the squared RKHS norm of this score:

$$I_j^K(G_j)^2 = \|S_n(G_j)\|_{\mathcal{H}}^2.$$

By the reproducing property, we obtain

$$\begin{aligned} I_j^K(G_j)^2 &= \left\langle \frac{1}{n}\sum_{i=1}^{n} K^1_{t_i(G_j)}, \frac{1}{n}\sum_{\ell=1}^{n} K^1_{t_\ell(G_j)} \right\rangle_{\mathcal{H}} \\ &= \frac{1}{n^2}\sum_{i=1}^{n}\sum_{\ell=1}^{n} K^1(t_i(G_j), t_\ell(G_j)) \,. \end{aligned}$$

Because the alternative space requires centered main effects in the two mirror branches and allows arbitrary dependence on W_j, the alternative kernel factorizes as

$$K^1((u, v, w), (u', v', w')) = K_U^1(u, u')K_V^1(v, v')K_W(w, w').$$

Since centering the U- and V-branch kernels is achieved by the centering matrix $H_n = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, the Gram matrix of K_U^1 and K_V^1 are $H_n K_U(G_j) H_n$ and $H_n K_V(G_j) H_n$, respectively. The W_j-kernel matrix is denoted by K_W. Substituting the factorized kernel into the double sum gives

$$\begin{aligned} I_j^K(G_j)^2 &= \frac{1}{n^2}\sum_{i=1}^{n}\sum_{\ell=1}^{n} \big[H_n K_U(G_j) H_n\big]_{i\ell} \big[H_n K_V(G_j) H_n\big]_{i\ell} \big[K_W\big]_{i\ell} \\ &= \frac{1}{n^2}\left[(H_n K_U(G_j) H_n) \circ (H_n K_V(G_j) H_n) \circ K_W\right]_{++} . \end{aligned}$$

This is exactly the stated grouped Neural Gaussian Mirror objective, thus completes the proof. □

C.10 Proof of Proposition 6

Proposition 6. *Under linear kernels and ignoring the W_j-kernel weight, the objective satisfies*

$$I_j^K(G_j)^2 \propto \left\| \frac{1}{n} \big(H_n U_j(G_j)\big)^\top \big(H_n V_j(G_j)\big) \right\|_F^2,$$

which reduces to the block-orthogonality target in the linear Group Gaussian Mirror.

Proof. We consider the linear-kernel special case and ignore the W_j-kernel weight, as stated in the proposition. Under linear kernels on the two mirror branches, we have

$$K_U(G_j) = U_j(G_j)U_j(G_j)^\top, \qquad K_V(G_j) = V_j(G_j)V_j(G_j)^\top.$$

Since H_n is symmetric and idempotent, define the centered mirror matrices $U_{j,c}(G_j) = H_n U_j(G_j)$ and $V_{j,c}(G_j) = H_n V_j(G_j)$, we then have

$$H_n K_U(G_j) H_n = H_n U_j(G_j) U_j(G_j)^\top H_n = U_{j,c}(G_j) U_{j,c}(G_j)^\top,$$

$$H_n K_V(G_j) H_n = V_{j,c}(G_j) V_{j,c}(G_j)^\top.$$

Ignoring the W_j-kernel weight, the grouped neural mirror objective is proportional to

$$\left[(H_n K_U(G_j) H_n) \circ (H_n K_V(G_j) H_n) \right]_{++}.$$

Substituting the two centered Gram matrices, we obtain

$$\begin{aligned} &\left[(H_n K_U(G_j) H_n) \circ (H_n K_V(G_j) H_n) \right]_{++} \\ &\quad = \sum_{i=1}^n \sum_{\ell=1}^n \big(u_{j,c,i}(G_j)^\top u_{j,c,\ell}(G_j) \big) \big(v_{j,c,i}(G_j)^\top v_{j,c,\ell}(G_j) \big), \end{aligned}$$

where $u_{j,c,i}(G_j)$ and $v_{j,c,i}(G_j)$ denote the i-th rows of $U_{j,c}(G_j)$ and $V_{j,c}(G_j)$, respectively. The preceding double sum can be rewritten as a trace:

$$\begin{aligned} &\sum_{i=1}^n \sum_{\ell=1}^n \big(u_{j,c,i}(G_j)^\top u_{j,c,\ell}(G_j) \big) \big(v_{j,c,i}(G_j)^\top v_{j,c,\ell}(G_j) \big) \\ &\quad = \operatorname{tr}\big(U_{j,c}(G_j) U_{j,c}(G_j)^\top V_{j,c}(G_j) V_{j,c}(G_j)^\top \big) \\ &\quad = \operatorname{tr}\big(U_{j,c}(G_j)^\top V_{j,c}(G_j) V_{j,c}(G_j)^\top U_{j,c}(G_j) \big) \\ &\quad = \big\| U_{j,c}(G_j)^\top V_{j,c}(G_j) \big\|_F^2. \end{aligned}$$

Therefore, we obtain

$$I_j^K(G_j)^2 \propto \frac{1}{n^2} \big\| U_{j,c}(G_j)^\top V_{j,c}(G_j) \big\|_F^2 = \left\| \frac{1}{n} \big(H_n U_j(G_j)\big)^\top \big(H_n V_j(G_j)\big) \right\|_F^2.$$

The matrix $\frac{1}{n}\big(H_n U_j(G_j)\big)^\top \big(H_n V_j(G_j)\big)$ is the centered empirical block cross-covariance between the two mirror branches. Hence, in the linear-kernel case, minimizing $I_j^K(G_j)^2$ is equivalent to driving this block cross-covariance toward zero, which is the block-orthogonality target in linear Group Gaussian Mirror. This completes the proof. □

C.11 Proof of Theorem 5

Theorem 5. *For any grouped feature j, let $Z_j \in \mathbb{R}^{n\times m}$ independently generated and satisfy $Z_j \overset{d}{=} -Z_j$ and $Z_j \perp (X, \boldsymbol{y})$. Assume $\widetilde{Z}_{j,\lambda}$ is odd and $\widehat{G}_j$ is even in Z_j, and the downstream training-and-attribution procedure is swap-equivariant in distribution under exchanging the two mirror branches. Then the mirror outputs $O_j^\pm$ are exchangeable in distribution, namely, $(O_j^+, O_j^-) \overset{d}{=} (O_j^-, O_j^+)$.*

Proof. Fix a grouped feature j. Let $Z_j \in \mathbb{R}^{n\times m}$ satisfy $Z_j \overset{d}{=} -Z_j$ and $Z_j \perp (X, \boldsymbol{y})$ as described. By assumption, the projected noise block $\widetilde{Z}_{j,\lambda}$ is odd in Z_j, and the learned perturbation matrix $\widehat{G}_j$ is even in Z_j, namely

$$\widetilde{Z}_{j,\lambda}(-Z_j) = -\widetilde{Z}_{j,\lambda}(Z_j), \qquad \widehat{G}_j(-Z_j) = \widehat{G}_j(Z_j).$$

The evenness of $\widehat{G}_j$ is supported by the symmetry of the kernel objective. Since changing Z_j to $-Z_j$ exchanges U_j and V_j, and the objective is symmetric in the two branches, we have

$$I_j^K(G; -Z_j)^2 = I_j^K(G; Z_j)^2.$$

Thus, the two optimization problems have identical argmin sets. The evenness of the selected $\widehat{G}_j$ then follows if the minimizer is unique or a sign-invariant optimization procedure is used, with identical initialization and coupled optimization randomness.

The final mirror branches are constructed as $U_j = X_j + \widetilde{Z}_{j,\lambda}\widehat{G}_j$ and $V_j = X_j - \widetilde{Z}_{j,\lambda}\widehat{G}_j$. We first show that flipping the mirror noise swaps the two branches. Under the transformation $Z_j \mapsto -Z_j$,

$$\begin{aligned} U_j(-Z_j) &= X_j + \widetilde{Z}_{j,\lambda}(-Z_j)\widehat{G}_j(-Z_j) \\ &= X_j - \widetilde{Z}_{j,\lambda}(Z_j)\widehat{G}_j(Z_j) \\ &= V_j(Z_j). \end{aligned}$$

$$\begin{aligned} V_j(-Z_j) &= X_j - \widetilde{Z}_{j,\lambda}(-Z_j)\widehat{G}_j(-Z_j) \\ &= X_j + \widetilde{Z}_{j,\lambda}(Z_j)\widehat{G}_j(Z_j) \\ &= U_j(Z_j). \end{aligned}$$

Thus the sign flip of the auxiliary mirror noise does not change the construction except for exchanging the plus and minus mirror branches. Let $\Psi(y, W_j, U_j, V_j) = (O_j^+, O_j^-)$ denote the downstream training-and-attribution map. The assumed swap-equivariance requires that exchanging the two mirror branches, together with their corresponding initialization, optimization randomness, and SHAP sampling randomness, exchanges the resulting importance scores. Identical preprocessing, architecture, and optimization rules for both branches, together with an exchangeable initialization distribution, support this property in distribution. Under the assumed equivariance, we have

$$\Psi(y, W_j, V_j, U_j) = (O_j^-, O_j^+),$$

with the equality interpreted under consistently coupled randomness.

Combining this with the branch-swapping identity above gives

$$(O_j^+, O_j^-)(-Z_j) = (O_j^-, O_j^+)(Z_j).$$

Because $Z_j \overset{d}{=} -Z_j$, the joint distribution of the full construction under Z_j is the same as under $-Z_j$, except for the induced exchange of the two mirror branches. The construction of $\widetilde{Z}_{j,\lambda}$ and $\widehat{G}_j$ is label-free, and Z_j is independent of $(X, \boldsymbol{y})$. Therefore, under the null,

$$(O_j^+, O_j^-) \overset{d}{=} (O_j^-, O_j^+).$$

This completes the proof. □

Remark (Null symmetry of mirror statistics). The branch exchangeability in Theorem 5 does not directly imply the null symmetry of the signed-inner-product mirror statistic, since $M_j(O_j^+, O_j^-) = M_j(O_j^-, O_j^+)$. An additional sufficient condition is one-branch sign symmetry:

$$(O_j^+, O_j^-) \overset{d}{=} (O_j^+, -O_j^-),$$

which implies $M_j \overset{d}{=} -M_j$. Asymptotically, this condition can follow when systematic null attributions are negligible relative to centered estimation errors and the errors satisfy joint one-branch sign symmetry, for example under a centered joint Gaussian limit with vanishing cross-covariance. The additional challenge of controlling systematic null attribution is further discussed in Appendix D.

D Multiplicity Concern and Practical Implications

An important concern for our framework is predictive and explanation multiplicity: multiple models with similar predictive risk may rely on different feature relationships and yield different attributions. This is related to underspecification [16] and attribution inconsistency within the Rashomon set [35]. In our setting, some near-optimal networks may recover the intended relationship, while others may achieve similar prediction accuracy through correlated null features or feature combinations, corresponding to shortcut learning [21].

To formalize this issue, define the Rashomon set

$$\mathcal{R}_n(\eta_n) = \left\{ f \in \mathcal{F}_n : R(f) \leq \inf_{g \in \mathcal{F}_n} R(g) + \eta_n \right\},$$

and let

$$\mu_j^{\pm}(f) = \mathbb{E}_{Q_j}\left[\psi_j^{\pm}(\mathbf{X}; f)\right]$$

denote the population aggregated Permutation-SHAP derivative for the two mirror branches. A stronger model-class-level condition is null-attribution identifiability:

$$\sup_{f \in \mathcal{R}_n(\eta_n)} \max_{j \in S_0} \left\{ \left\|\mu_j^+(f)\right\|_2 + \left\|\mu_j^-(f)\right\|_2 \right\} \longrightarrow 0. \tag{R1}$$

Condition (R1) requires all sufficiently predictive models, rather than a single fitted model, to assign negligible attribution to null features. This is stronger than predictive consistency, since similar prediction error does not imply agreement on feature usage.

However, (R1) is stronger than necessary for one realized PSGM model. For the fitted predictor $\widehat{f}_n$, the direct requirement is asymptotic null-attribution consistency. Writing

$$\widehat{\psi}_j^{\pm} = \underbrace{\mu_j^{\pm}(\widehat{f}_n)}_{\text{systematic fitted-model attribution}} + \underbrace{r_n \varepsilon_j^{\pm}}_{\text{centered statistical error}} + o_p(r_n),$$

we require

$$\max_{j \in S_0} \frac{\left\|\mu_j^+(\widehat{f}_n)\right\|_2 + \left\|\mu_j^-(\widehat{f}_n)\right\|_2}{r_n} \xrightarrow{p} 0. \tag{R2}$$

Thus, finite-sample or optimization-induced null dependencies are allowed, but they must vanish relative to attribution-estimation noise. Persistent directional attribution to a null group represents systematic shortcut learning and can invalidate null mirror symmetry.

This distinction clarifies the scope of PSGM's FDR guarantee. The attribution layer is model-conditional: after fitting a predictor, Permutation SHAP characterizes how that predictor uses its inputs. If a fitted network relies on a feature satisfying

$$Y \perp X_j \mid X_{-j},$$

the feature remains statistically null but may be important for the fitted model. Its attribution can therefore be faithful to the model while misleading about the underlying data-generating mechanism. Condition (R2) connects these two notions.

More broadly, FDR procedures control errors relative to a specified statistical null under assumptions such as exchangeability or symmetry. They do not generally resolve which model in a Rashomon set represents the intended mechanism. Therefore, PSGM does not eliminate predictive multiplicity; its neural guarantee is conditional on sufficiently small systematic null attribution of the fitted model.

A possible extension is to define a model-class-level attribution target. [20] proposed model class reliance, which evaluates variable importance over well-performing models, while [35] study attribution consensus within the Rashomon set. This would directly address predictive multiplicity but requires a different inferential framework and new FDR analysis.

Some practical implications for model training. Although generic training cannot guarantee null-attribution identifiability, the preceding analysis provides several practical principles. First, two mirror branches should receive identical preprocessing, architecture, regularization, initialization distributions, and optimization rules, so that the training procedure does not introduce an artificial plus/minus preference. Second, weight decay, dropout, early stopping, and restricted network capacity can reduce the ability of an overparameterized model to retain weak shortcut dependencies, although they do not guarantee that the selected model is the correct member of the Rashomon set. Third, models with comparable validation error can be compared across random initializations, with preference given to solutions whose attributions are more stable under retraining and mirror-branch relabeling.

When prior information about invalid dependencies is available, a more direct possibility is attribution regularization. For example, [46] penalize model gradients on features that should not determine the prediction, thereby training models to be "right for the right reasons." When multiple environments or predefined subgroups are available, invariant or group-robust objectives can similarly discourage relationships that are predictive only in one environment or subgroup. These techniques are possible mitigations of shortcut learning, rather than replacements for the explicit assumptions in our FDR theorem.

E Sensitivity Analysis for Monte Carlo Permutation SHAP

Here we conduct experiments to support the implementation choice in Section 4.1. In PSGM, the exact group-level Permutation SHAP value is approximated by Monte Carlo permutation sampling before estimating the smoothed SHAP derivative. We therefore vary the number of permutation draws from 3 to 316 and evaluate whether the resulting FDR and power are sensitive to this approximation. The results show that increasing the number of draws has only a limited effect on selection performance, and that 10 permutation draws are sufficient for stable results in our simulations. We use this value in the main neural experiments.

Table 8: FDR and power of PSGM under different numbers of Monte Carlo permutation draws

Setting (p*m, n)	Model	Signal Strength	MC3		MC10		MC32		MC100		MC316	
			FDR	Power	FDR	Power	FDR	Power	FDR	Power	FDR	Power
$p*m=1500$ $n=2500$	LSTM	Weak	0.182	**0.896**	0.161	0.879	0.143	0.844	**0.123**	0.812	0.127	0.851
		Strong	0.148	0.916	**0.145**	**0.958**	0.164	0.937	0.167	0.943	0.174	0.957
	GRU	Weak	0.132	**0.934**	0.133	0.931	0.099	0.897	**0.086**	0.874	0.097	0.904
		Strong	0.113	0.951	0.162	0.979	**0.102**	0.985	0.185	0.982	0.162	**0.994**
	Transformer	Weak	0.073	**0.855**	0.068	0.846	0.052	0.812	**0.029**	0.811	0.056	0.813
		Strong	0.106	0.855	0.116	0.887	0.083	0.893	**0.082**	**0.895**	0.128	0.893
$p*m=3500$ $n=2500$	LSTM	Weak	**0.111**	0.615	0.116	**0.625**	0.119	0.618	0.149	0.617	0.168	0.616
		Strong	0.152	0.865	0.152	0.886	0.145	0.895	0.147	**0.899**	**0.144**	0.893
	GRU	Weak	0.063	0.798	0.062	0.817	**0.041**	**0.820**	0.079	0.806	0.089	0.812
		Strong	0.235	0.884	**0.223**	0.919	0.224	0.911	0.243	**0.931**	0.251	0.926
	Transformer	Weak	**0.104**	0.492	0.141	0.491	0.112	0.503	0.155	**0.510**	0.128	0.501
		Strong	0.237	0.874	0.283	0.900	0.254	0.889	0.208	**0.921**	**0.196**	0.892

F Convergence Analysis for Kernel-Based Objective Optimization

Here we conduct experiments to support the perturbation-matrix optimization step introduced in Section 2.2 and Section 4. In the grouped neural mirror construction, the scalar perturbation in Neural Gaussian Mirror is replaced by a matrix G_j, which is optimized by minimizing the kernel-based conditional-dependence objective $I_j^K(G_j)^2$. We track the standardized objective value during optimization to verify that this step is numerically stable and does not require heavy tuning. The results show that initializing $G_j = I_m$ and using a fixed learning rate and number of epochs yields stable convergence, motivating the fixed choice used in the later neural experiments.

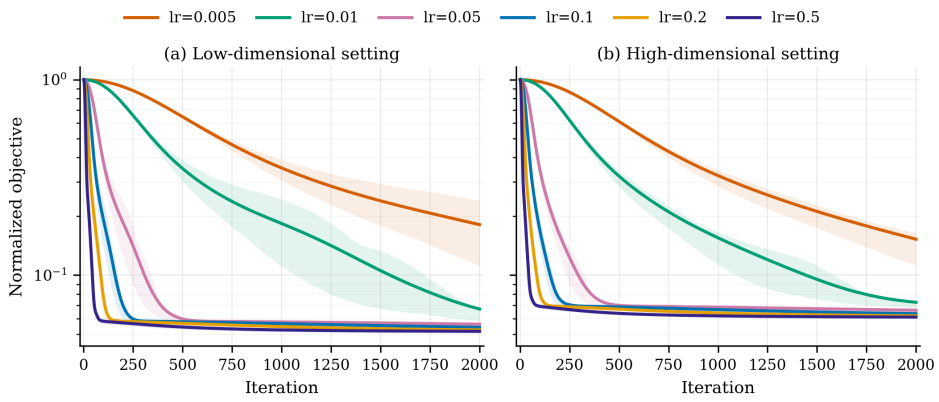


Figure 2: Convergence curve of the kernel-based objective optimization

G Complexity and Scale Sensitivity Analysis for PSGM

Here we show runtime/memory complexity and performance sensitivity analysis of PSGM to different scales of p, n, m in both low- and high-dimensional settings. First we scale $((p * m), n)$ by (2*2) for both cases and record average performance and runtime/memory all on same the hardware we used in the Numerical Simulation section. Empirically we find the performance degrades high-dim case with larger p and n. However it is more like a model predictive multiplicity issue introduced in see Appendix D. We also find the whole algorithm is 6-8 times slower when (p*m*n) is 4 times larger. Also as we mentioned, since the G_j optimization step is the slowest, this can effectively accelerated by increasing the learning rate.

Second, we add sensitivity analysis to scales of m only. We fix p and n and test $m = 3$ and $m = 15$. Empirically as we stated in the paper, the performance degrades larger m. This is also mostly due to model predictive multiplicity issue. We also compare runtime and memory with the two baselines, demonstrating that our method achieves superior performance without adding too much algorithm complexity.

Table 9: Complexity and scale sensitivity analysis for PSGM

Experiment	Method	p	m	n	FDR	Power	Runtime (min)	Peak memory (GB)
Scale–LD	PSGM	300	5	2500	0.063	1.000	3.661	5.854
Scale–LD	PSGM	600	5	5000	0.151	1.000	28.815	11.746
Scale–LD	PSGM	1200	5	10000	0.218	1.000	153.544	24.093
Scale–HD	PSGM	700	5	2500	0.091	0.667	5.972	13.869
Scale–HD	PSGM	1400	5	5000	0.167	1.000	42.097	27.378
Scale–HD	PSGM	2800	5	10000	0.583	1.000	340.836	56.162
m sensitivity	PSGM	300	3	2500	0.000	1.000	3.452	3.561
m sensitivity	PSGM	300	5	2500	0.063	1.000	3.661	5.854
m sensitivity	PSGM	300	15	2500	0.417	0.467	3.912	17.173

H Additional Linear Simulation Results

This appendix reports the performance and average runtime of another baseline, Group Knockoff [15], under the same experiment setting in Section 5.1. We only experiment Group Knockoff on limited within-group AR coefficient and among-group correlation settings because the method is significantly slower. We found our method much faster yet performs similar.

This appendix also reports additional linear simulation results under a block-correlated Gaussian grouped design. This design complements the AR grouped design in Section 5.1: instead of generating each grouped feature from lagged values of a latent AR process, we generate a static Gaussian design with controlled within-group and between-group correlations. This allows us to check whether the performance of GGM depends on the temporal AR construction used in the main linear simulations.

In this design, each row of X is sampled from a multivariate Gaussian distribution. For each grouped feature X_j, the m sub-features have within-group correlation $\rho_w \in \{0.1, 0.5, 0.9\}$. Columns from different grouped features have among-group correlation $\rho_b = \alpha\rho_w$, with $\alpha \in \{0.2, 0.8\}$. For each non-null grouped feature, the sub-feature weights are sampled independently from $N(0, \sigma_\beta^2)$, rather than following the decreasing lag profile used in the AR grouped design. Table 11 reports the averaged grouped-feature FDR and power over 50 repetitions. The results show that GGM maintains stable FDR control and high power under this alternative block-correlated Gaussian design.

Table 10: Statistical performance and computational cost of Group Knockoff

Within-group AR	Among-Group corr.	p	m	n	FDR	Power	Runtime (min)
0.1	0.2	300	5	2500	0.020	1.000	294.564
0.1	0.8	300	5	2500	0.103	0.995	344.014
0.5	0.2	300	5	2500	0.073	1.000	320.738
0.5	0.8	300	5	2500	0.045	1.000	342.865

Table 11: Linear simulation results under the block-correlated Gaussian grouped design

Setting (p*m, n)	Among-Group Corr	In-Group Corr	GGM-OLS		GGM-GLS		Partial-F BH		Group Lasso	
			FDR	Power	FDR	Power	FDR	Power	FDR	Power
$p*m=1500$, $n=2500$	0.2	0.1	**0.076**	1.000	0.077	1.000	0.090	1.000	0.316	1.000
		0.5	0.088	1.000	0.091	1.000	**0.086**	1.000	0.402	1.000
		0.9	0.104	0.997	**0.078**	0.997	0.087	0.999	0.453	0.993
	0.8	0.1	0.075	1.000	0.080	1.000	**0.061**	1.000	0.404	1.000
		0.5	0.111	1.000	0.124	1.000	**0.076**	1.000	0.381	1.000
		0.9	**0.084**	0.999	0.085	0.999	**0.084**	0.999	0.395	1.000
$p*m=3500$, $n=2500$	0.2	0.1	0.085	1.000	0.069	1.000	**0.065**	1.000	0.318	1.000
		0.5	**0.083**	1.000	0.097	1.000	0.094	1.000	0.442	1.000
		0.9	0.111	1.000	**0.065**	1.000	0.095	0.994	0.378	1.000
	0.8	0.1	0.067	1.000	0.074	0.995	**0.063**	1.000	0.415	0.997
		0.5	0.104	0.998	0.142	1.000	**0.058**	0.997	0.397	1.000
		0.9	**0.086**	0.996	0.091	1.000	0.088	1.000	0.383	0.999

I Additional Details for the CausalRivers Real Data Application

I.1 Dataset and Preprocessing

CausalRivers [51] is a large-scale real-world benchmark for causal discovery from river-discharge time series. In this paper, we use the East Germany subset, which contains 666 gauge stations observed at 15-minute resolution from January 2019 to December 2023, together with a directed upstream-to-downstream graph with 666 nodes and 651 edges. The dataset is useful for our setting because it combines realistic temporal dependence with physically meaningful graph information. However, the graph should only be treated as proxy relevance information in our regression problem. It should not be interpreted as a perfect statistical ground truth for our regression problem, since river discharge may also be affected by nonlinear flow, seasonality, weather-driven common shocks, variable causal lags, and unobserved confounding.

Let $R_{t,k}$ denote the discharge measurement at station k. We aggregate the raw 15-minute measurements to 3-hour averages, interpolate missing values, and use first differences $\Delta R_{t,k} = R_{t,k} - R_{t-1,k}$ to reduce persistent level effects and common low-frequency trends. We use the first 2400 differenced observations after January 1, 2021, and split them chronologically with a train-test ratio of $4:1$. For a target station j, the real response is $y_t^{\text{real}} = \Delta R_{t,j}$. For each candidate station k, we construct one grouped lag feature $X_{t,k} = (\Delta R_{t-1,k}, \Delta R_{t-2,k}, \Delta R_{t-3,k})$, so the selected unit is a station-level lag block rather than an individual lagged covariate. All covariates and responses are standardized using the training split.

I.2 Target and Grouped Feature Construction

We select six target stations using a fixed X-only screening rule. A target is retained if it has at least 10 graph-relevant candidate groups, 50 eligible graph-null groups, a low-dimensional feasible design for the linear benchmark, and an acceptable condition number. The selected target station IDs are

176, 166, 305, 304, 730, and 167. As shown in Table 12, each target design contains 64–66 station groups, including 14–16 graph-relevant groups and 50 sampled graph-null groups. Since each station group has three lags, this gives 192–198 scalar lagged covariates.

Table 12: Summary of the six CausalRivers target designs

Target ID	River system	# Direct	# 2nd	# Null	# Groups	# Lagged covariates	Cond. no.
176	Elbe	10	6	50	66	198	49.284
166	Elbe	5	11	50	66	198	64.500
305	Havel	6	9	50	65	195	24.152
304	Havel	4	10	50	64	192	24.750
730	Saale	6	8	50	64	192	25.281
167	Elbe	4	10	50	64	192	25.753

We define graph-proxy relevance using the directed river graph. Direct upstream stations are labeled as first-order relevant groups, and upstream stations of those direct upstream stations are labeled as second-order relevant groups. Graph-null groups are sampled from stations with no directed path relation to the target, excluding the target itself, its ancestors, and its descendants. Candidate stations with near-zero differenced variance are removed before fitting.

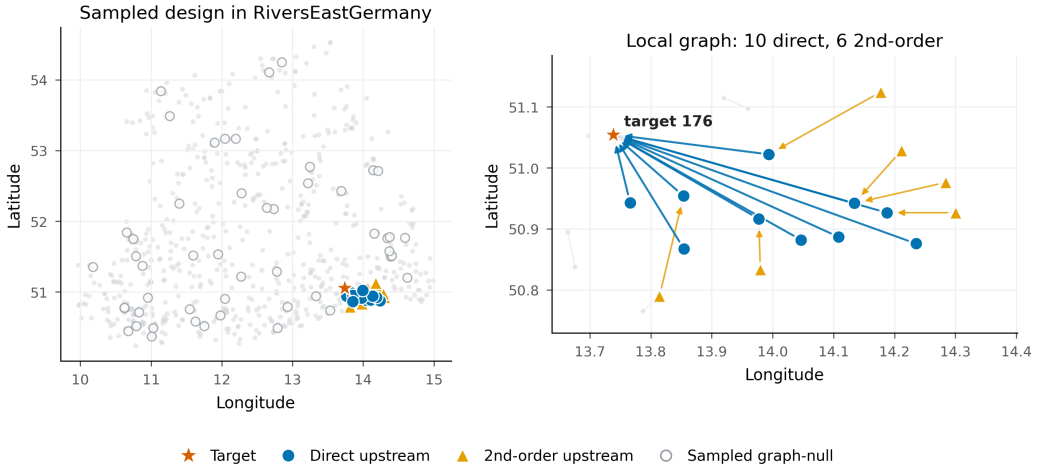


Figure 3: Graph-proxy relevance construction example for Target ID 176

I.3 Real and Semi-synthetic Responses

We evaluate both real and semi-synthetic responses, summarized in Table 13. The real response is $y_t^{\text{real}} = \Delta R_{t,j}$. Since the directed river graph does not fully determine the statistical null in our regression model, FDR and power for real y are reported as graph-proxy.

For semi-synthetic responses, we keep the real river covariates X fixed and define the true relevant set as $S_1 = \mathcal{S}_{\text{direct}} \cup \mathcal{S}_{\text{second}}$, where $\mathcal{S}_{\text{direct}}$ contains direct upstream groups and $\mathcal{S}_{\text{second}}$ contains second-order upstream groups. This preserves the real covariate dependence while allowing exact FDR and power evaluation.

For the linear semi-synthetic response, used in the linear benchmark, we generate

$$y_t^{\text{lin}} = \sum_{k \in S_1} X_{t,k}^\top \boldsymbol{\beta}_k + \varepsilon_t.$$

Direct upstream groups have coefficient scale 1.00, second-order upstream groups have coefficient scale 0.55, and the lag profile is $(1.00, 0.65, 0.35)$ up to an independent random sign per relevant group. Gaussian noise is calibrated to signal-to-noise ratio $= 2.0$.

For the nonlinear semi-synthetic response, used in the neural benchmark, we first compute $s_{t,k} = \boldsymbol{w}^\top X_{t,k}$, where $\boldsymbol{w}$ is an ℓ_2-normalized exponentially decaying lag-weight vector with larger weight

on recent lags. We then generate

$$\eta_t = \sum_{k \in \mathcal{S}_{\text{direct}}} \xi_k \tanh(s_{t,k}) + 0.55 \sum_{k \in \mathcal{S}_{\text{second}}} \xi_k \sin(s_{t,k}) + 0.20 \sum_{(k,k') \in \mathcal{P}_{\text{direct}}} \tanh(s_{t,k} s_{t,k'}/2),$$

where $\xi_k \in \{-1, 1\}$ and $\mathcal{P}_{\text{direct}}$ contains pairs of direct upstream groups. We set $y_t^{\text{nonlin}} = \eta_t + \varepsilon_t$, with Gaussian noise calibrated to signal-to-noise ratio $= 2.0$. The interaction term uses only direct upstream groups, so the true support remains unambiguous.

Table 13: Response types used in the CausalRivers application

Outcome	Covariates	Support label	Main purpose
Real y	Real river X	Graph-proxy	Graph-consistent real-data selection
Linear synthetic y	Real river X	Known support	Linear grouped-feature FDR evaluation
Nonlinear synthetic y	Real river X	Known support	Neural grouped-feature FDR evaluation

I.4 Methods and Evaluation Metrics

For the CausalRivers experiments, we use target FDR level $q = 0.2$ to allow a less conservative real-data selection. For the linear benchmark, we compare GGM with Partial-F BH and cross-validated Group Lasso. Partial-F BH computes one group-level partial-F p-value for each station block and applies the Benjamini–Hochberg procedure to the resulting grouped-feature p-values. Group Lasso is treated as a feature-selection baseline rather than an FDR-control method.

For the neural benchmark, we train LSTM, GRU, and Transformer encoder models on the same station-level lag-block design. The training split is further divided into fitting and validation blocks. We use hidden dimension 32, one layer, dropout 0.1, Adam optimizer with learning rate 10^{-3}, weight decay 10^{-4}, batch size 128, maximum 150 epochs, and early stopping patience 20. The Transformer uses 4 attention heads. For PSGM selection, we use grouped Permutation-SHAP importance with 10 permutation draws and train the selection model for 120 epochs.

For each method, we report FDR or proxy FDR, power or proxy power, selected-group rates for direct upstream, second-order upstream, and null groups, and test MSE. The columns $\%$ Direct, $\%$ 2nd, and $\%$ Null denote the fraction of groups selected within each category. We also report the full-model MSE and an oracle upstream-only MSE. For real y, this oracle is a graph-oracle model using direct and second-order upstream groups; for semi-synthetic y, it is the true-support oracle.

Table 14: Hyperparameters for CausalRivers experiments

Component	Value
Resampling frequency	3 hours
Look-back length	$L = 3$
Target FDR level	$q = 0.2$
Number of target stations	6
Graph-null groups per target-seed case	50
Train-test split	$4:1$, chronological
Neural hidden dimension	32
Neural layers	1
Dropout	0.1
Learning rate	10^{-3}
Weight decay	10^{-4}
Batch size	128
Maximum epochs	150
Early stopping patience	20
Transformer heads	4
PSGM permutation draws	10
PSGM selection epochs	120
Synthetic SNR	2.0

I.5 Linear CausalRivers Results

Table 15 reports the linear CausalRivers results for both real y and the linear semi-synthetic response. For real y, FDR and power are graph-proxy metrics based on the river graph labels. For semi-synthetic y, FDR and power are computed using the constructed true support. The table also reports category-wise selection rates and test MSE for selected, full, and oracle upstream-only models.

Table 15: Linear model results on the CausalRivers real data application

Outcome	Method	FDR/Proxy	Power/Proxy	% Direct	% 2nd	% Null	Sel. MSE	Full MSE	Oracle MSE
Real y	GGM	0.508	0.194	0.312	0.137	0.116	1.437	1.485	1.342
	Partial-F BH	0.576	0.342	0.431	0.293	0.148	1.302	1.485	1.342
	Group Lasso	**0.415**	**0.426**	0.490	0.390	0.238	**1.275**	1.485	1.342
Semi-synthetic y	GGM	0.219	0.922	0.974	0.885	0.144	0.400	0.394	0.358
	Partial-F BH	**0.127**	0.885	0.956	0.825	0.050	0.397	0.394	0.358
	Group Lasso	0.668	**0.976**	0.978	0.972	0.627	**0.381**	0.394	0.358

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